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(54) **SURGICAL IMPLANT DEVICES AND
METHODS FOR THEIR MANUFACTURE
AND USE**

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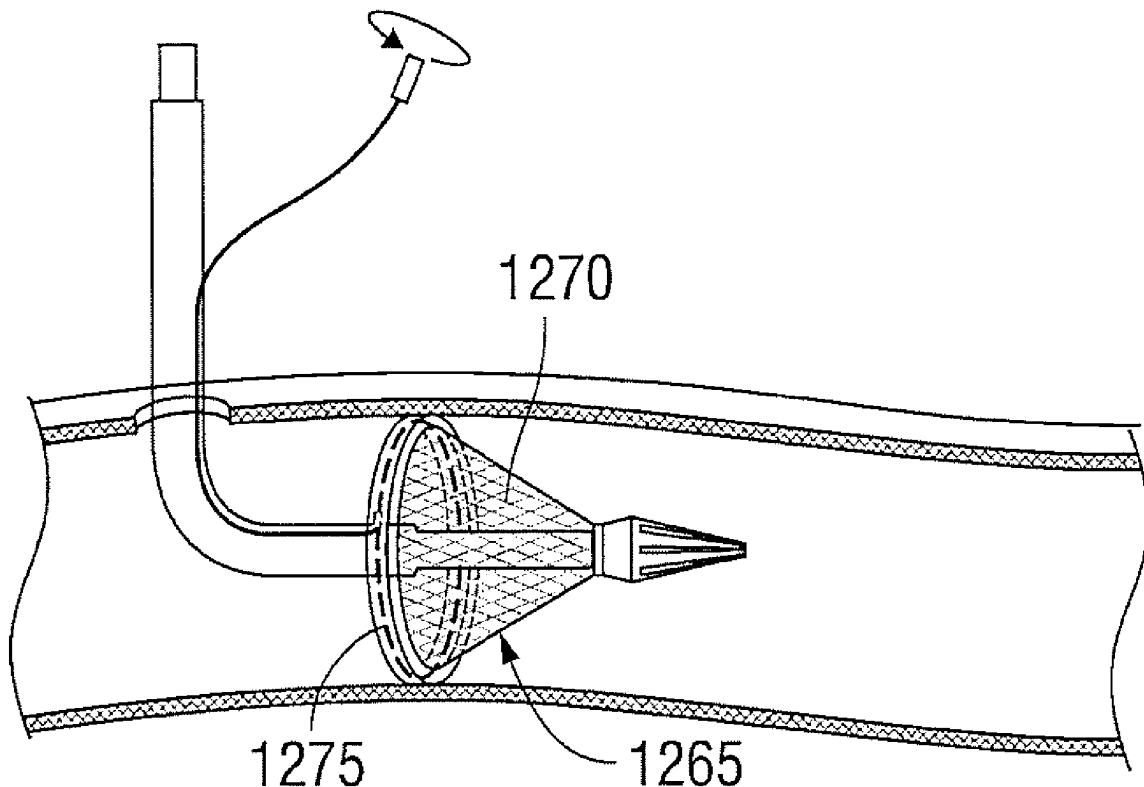
- (63) Continuation of application No. 18/086,059, filed on Dec. 21, 2022, now Pat. No. 12,329,630, which is a continuation of application No. 16/716,416, filed on Dec. 16, 2019, now Pat. No. 11,540,911, which is a continuation of application No. 15/448,417, filed on Mar. 2, 2017, now Pat. No. 10,507,097, which is a continuation of application No. 13/339,236, filed on Dec. 28, 2011, now Pat. No. 9,585,743.
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(2013.01); *A61F 2250/0098* (2013.01)

(57) **ABSTRACT**

An assembly includes an implant body and a delivery apparatus. The implant body includes a lattice structure, an adjustment member, and a control device. The lattice structure is circumferentially expandable and contractible. The adjustment member and the control device are coupled to the lattice structure. Actuating the adjustment member results in circumferential expansion or contraction of the lattice structure. The delivery apparatus includes a rotatable adjustment tool and a locking mechanism comprising a controllable catch coupled to an end portion of the rotatable adjustment tool. The rotatable adjustment tool is configured to actuate the adjustment member of the implant body upon rotation of the rotatable adjustment tool. The controllable catch of the locking mechanism is configured to selectively couple the rotatable adjustment tool to the adjustment member of the implant body.



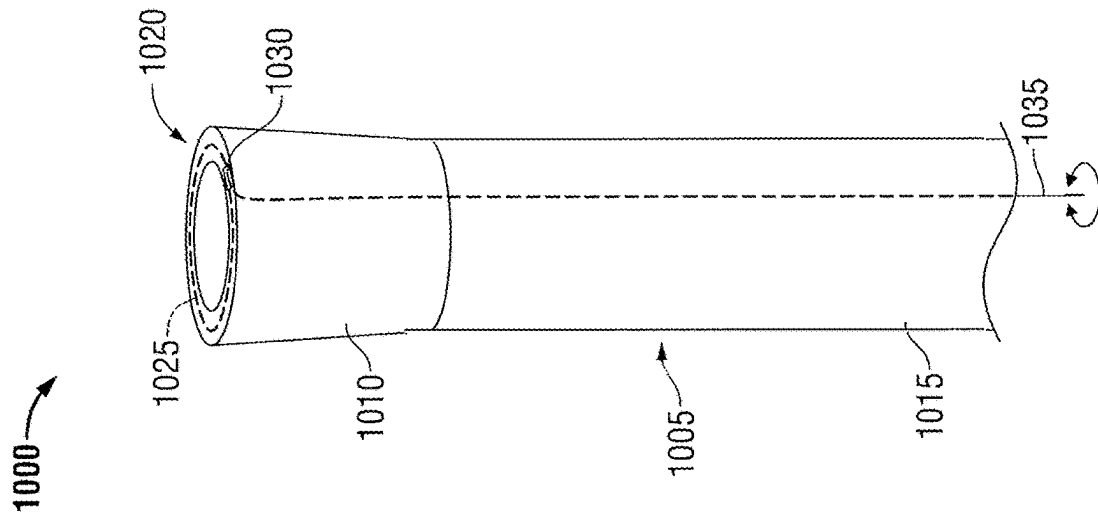


FIG. 1

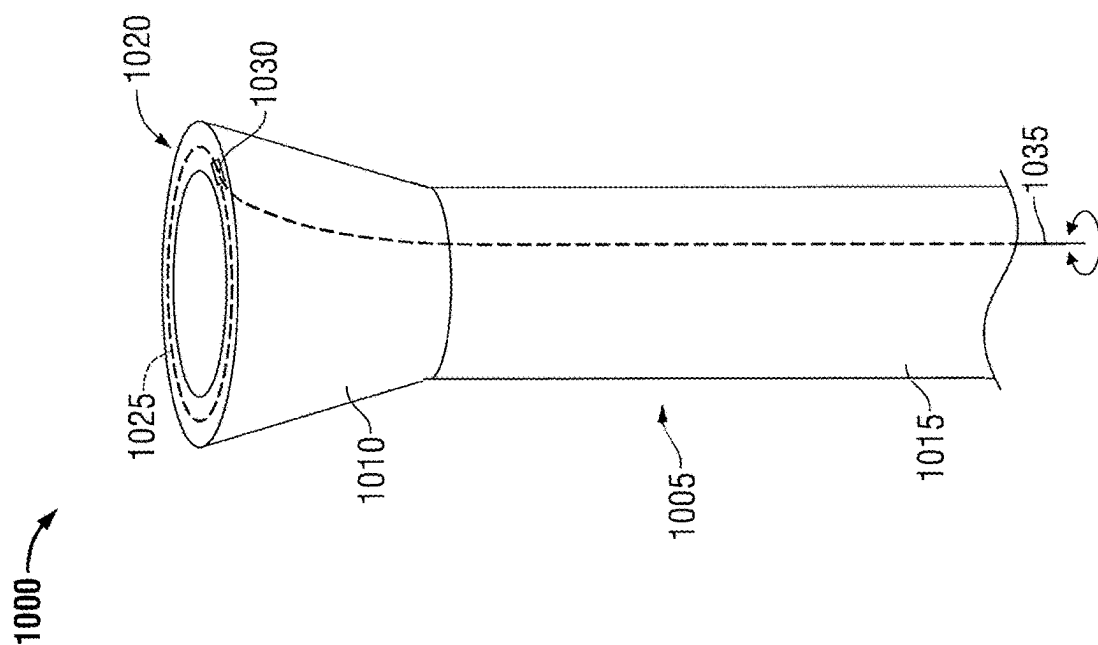


FIG. 2

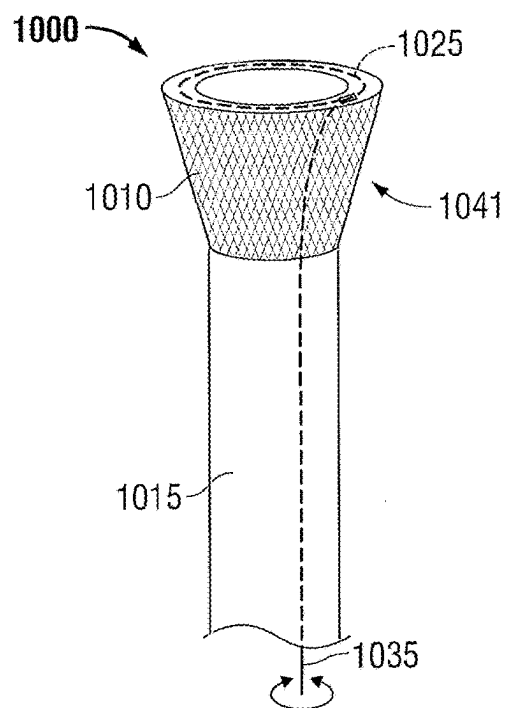


FIG. 3

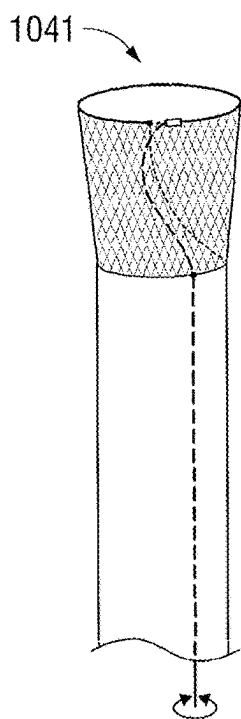


FIG. 4A

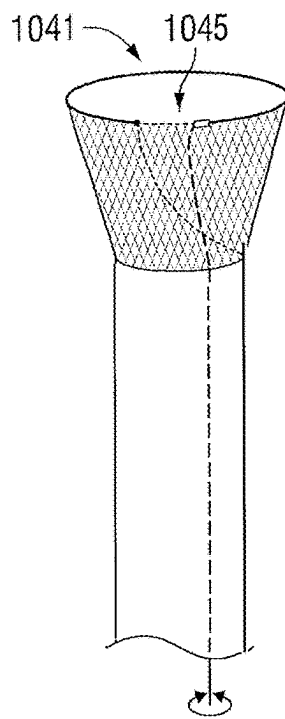


FIG. 4B

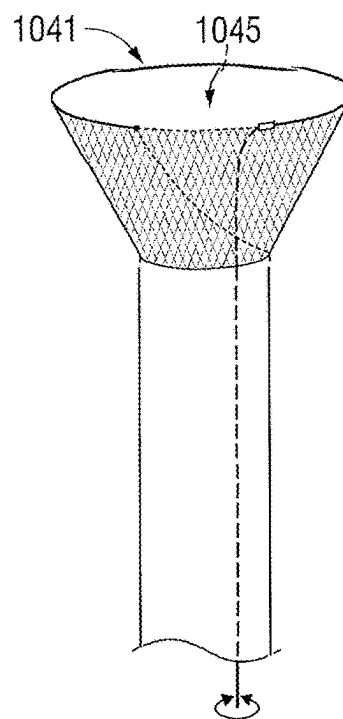
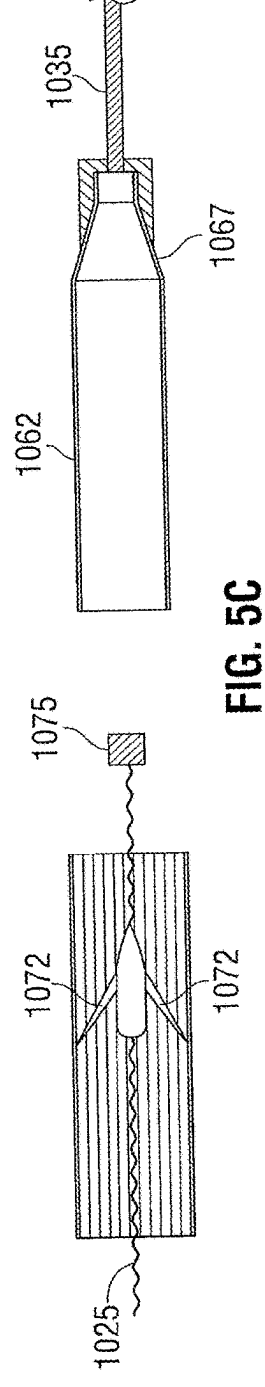
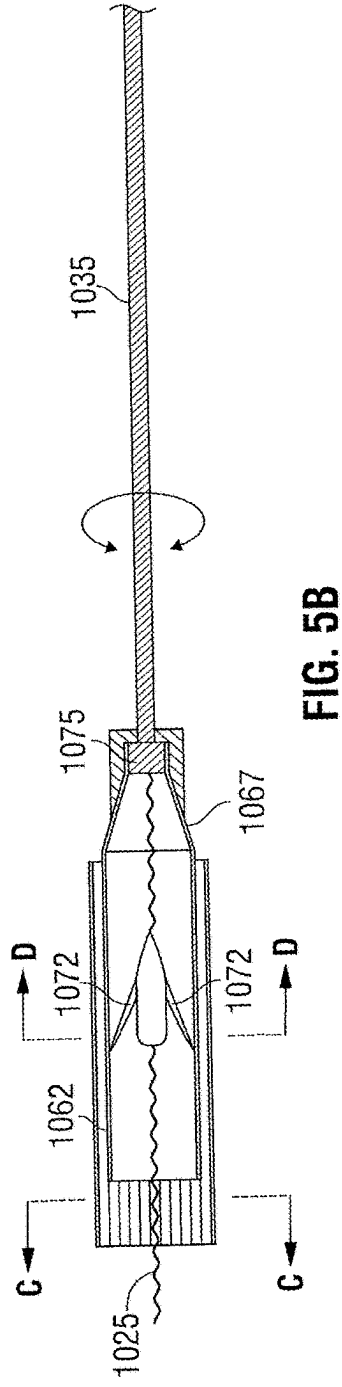
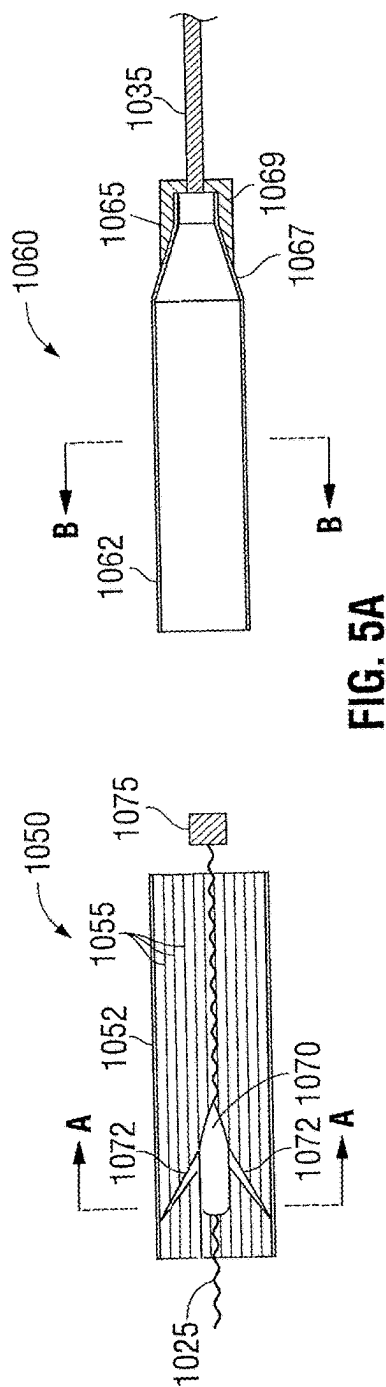


FIG. 4C



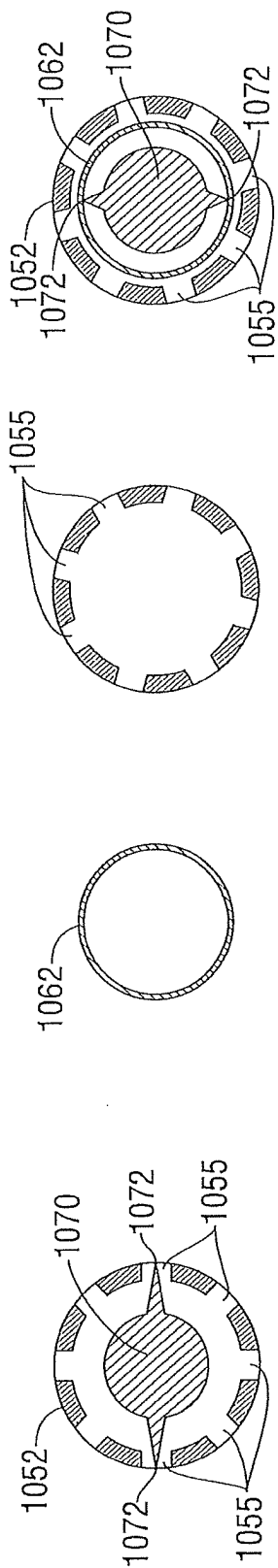


FIG. 6A

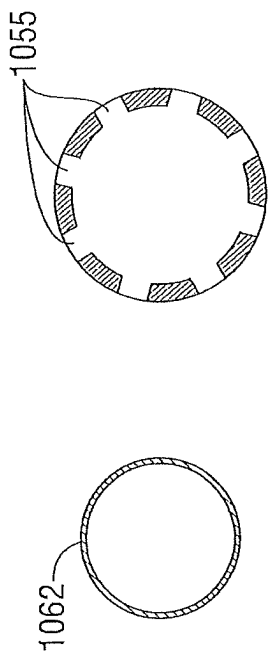


FIG. 6B

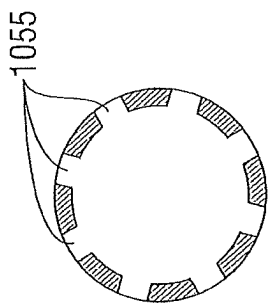


FIG. 6C

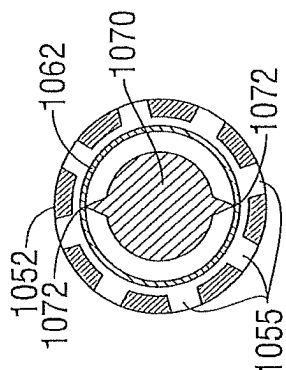


FIG. 6D

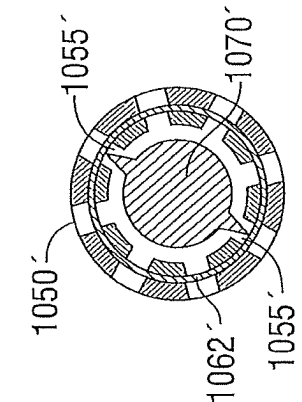


FIG. 6E

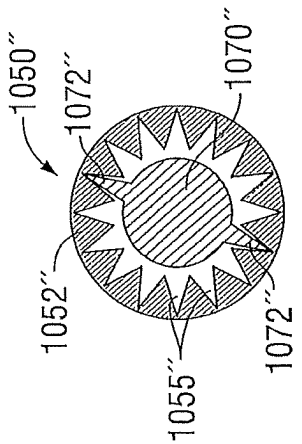


FIG. 6F

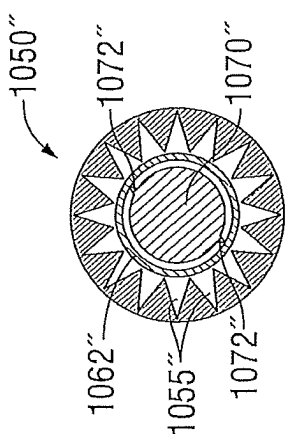


FIG. 6G

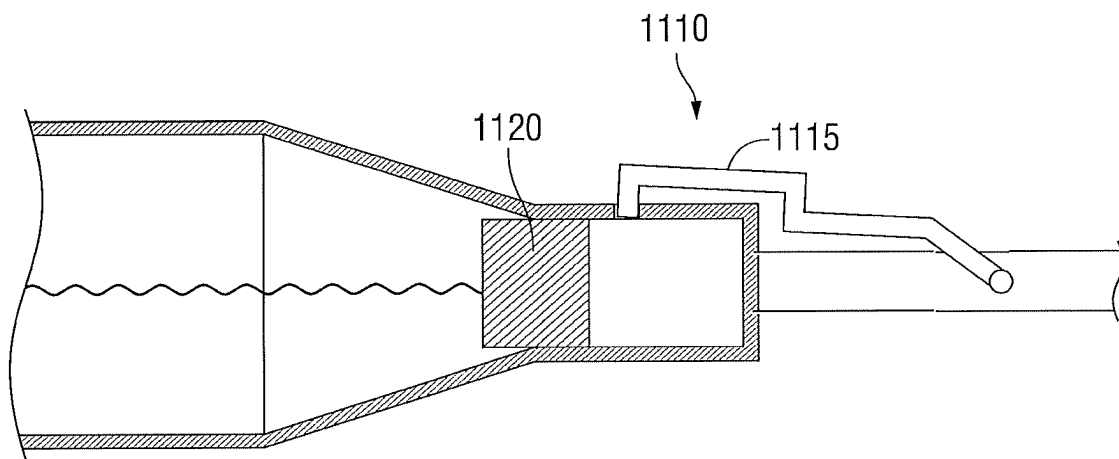


FIG. 7A

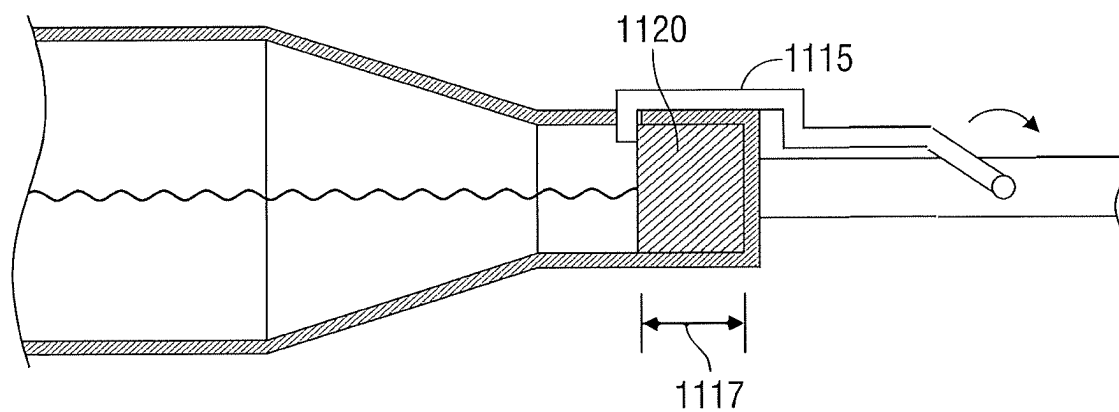


FIG. 7B

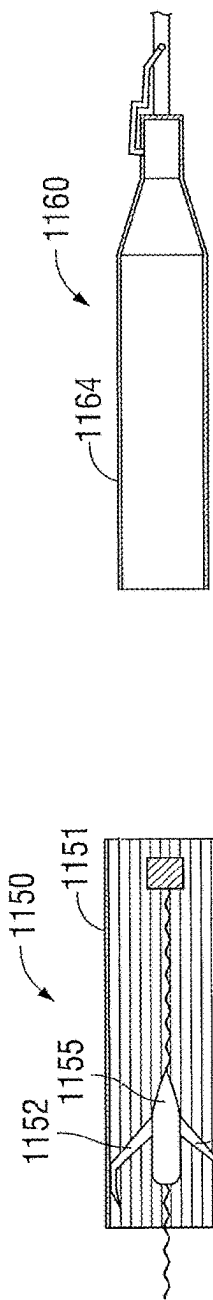


FIG. 8A

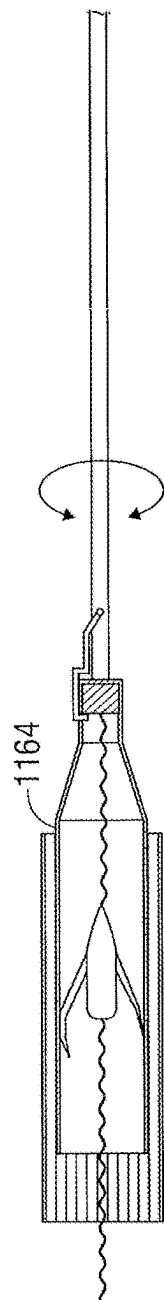


FIG. 8B

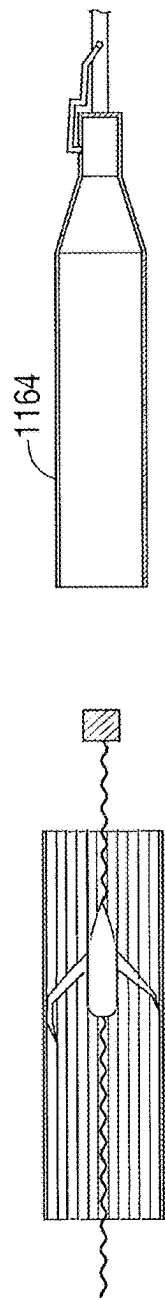


FIG. 8C

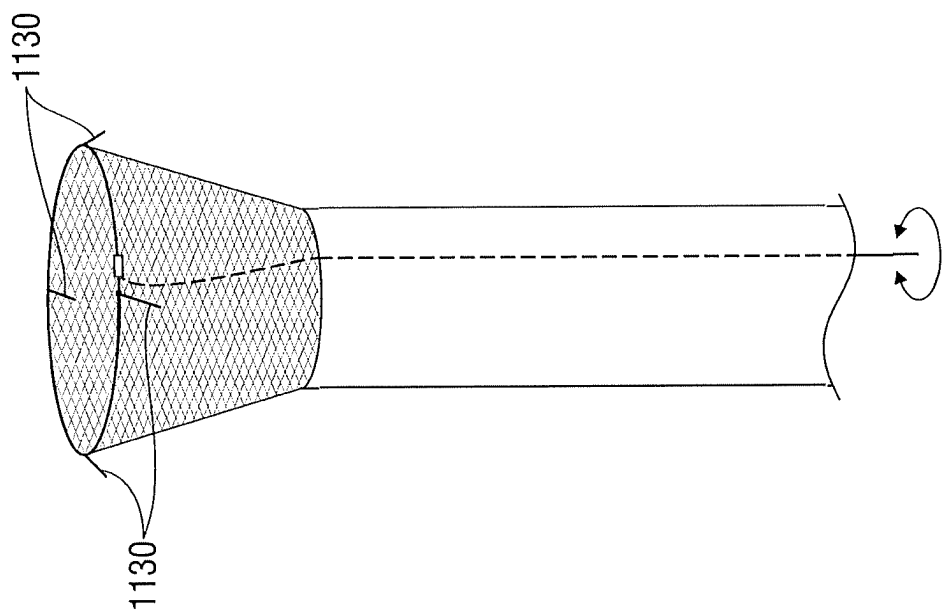


FIG. 9B

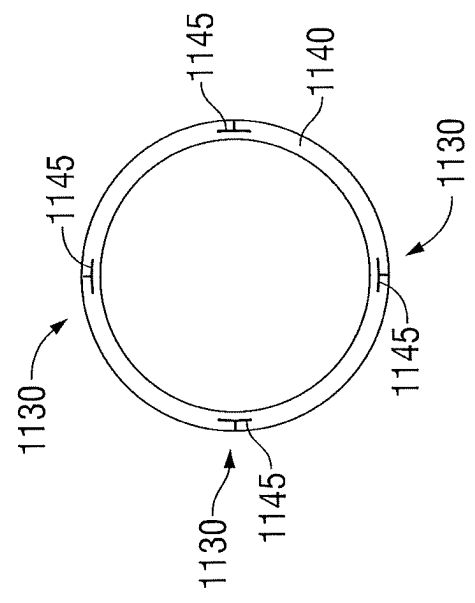


FIG. 9A

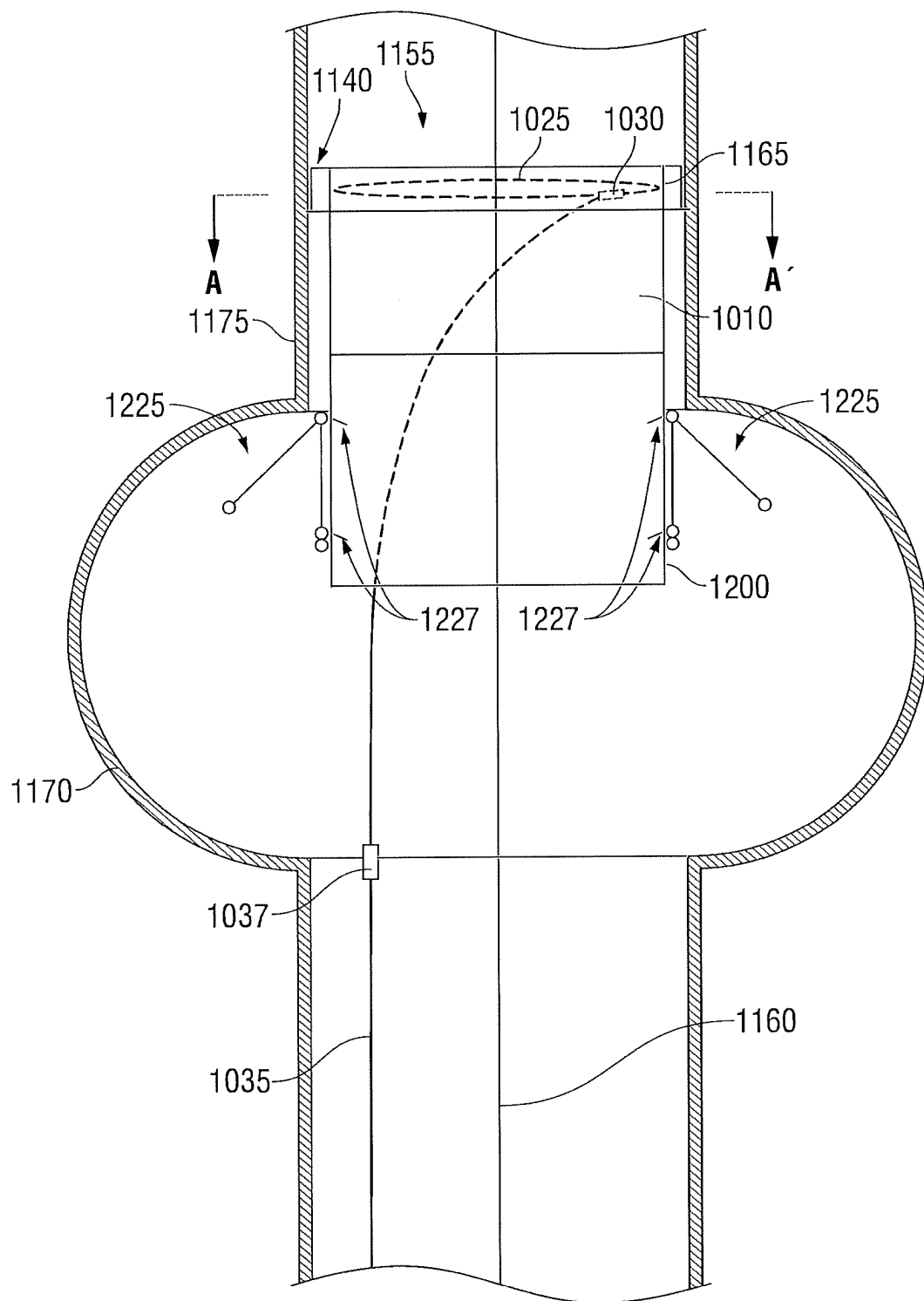


FIG. 10A

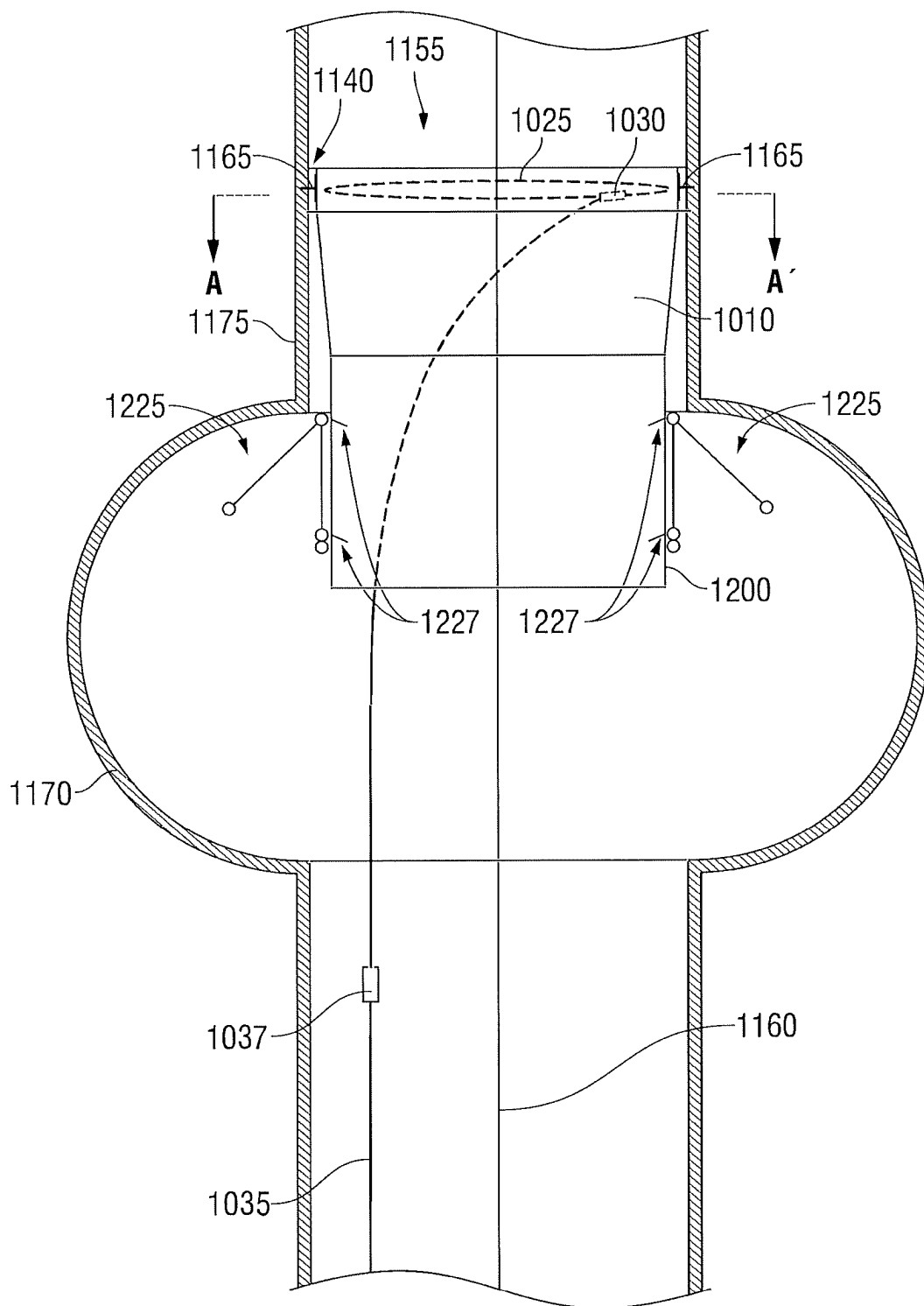


FIG. 11A

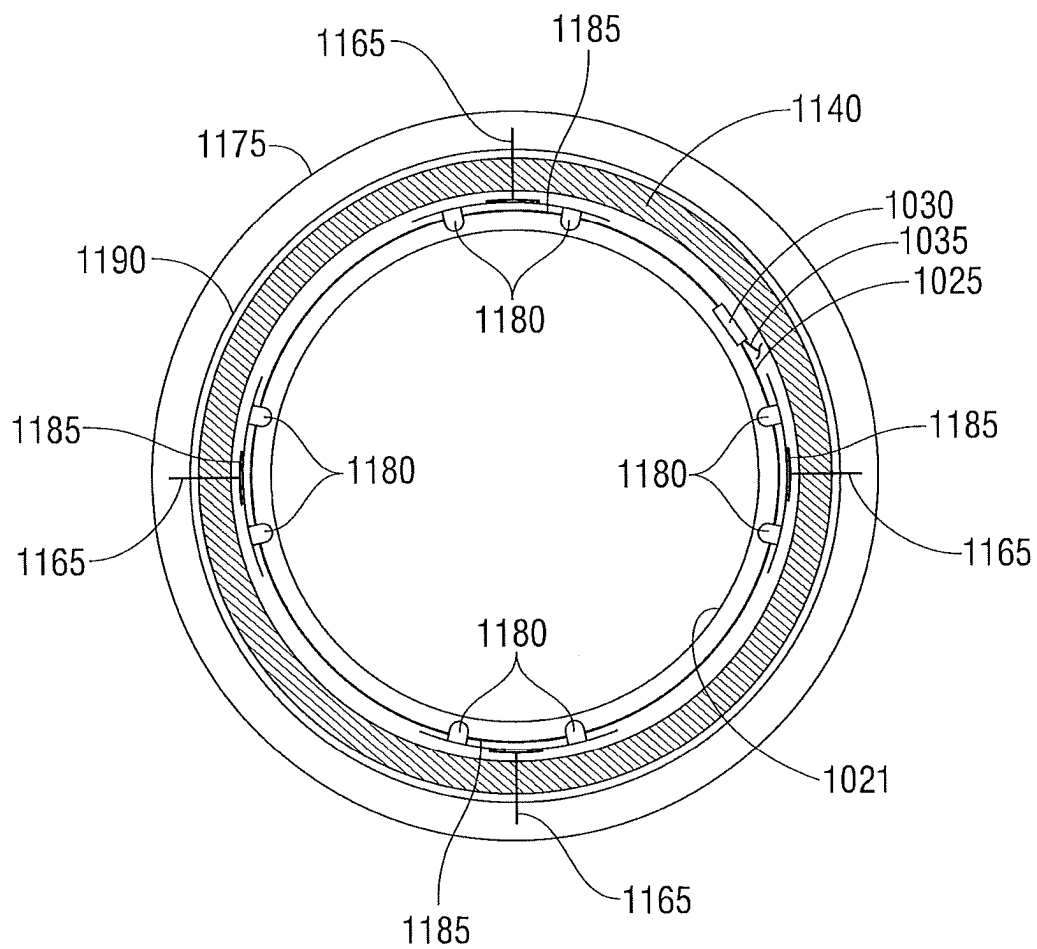


FIG. 11B

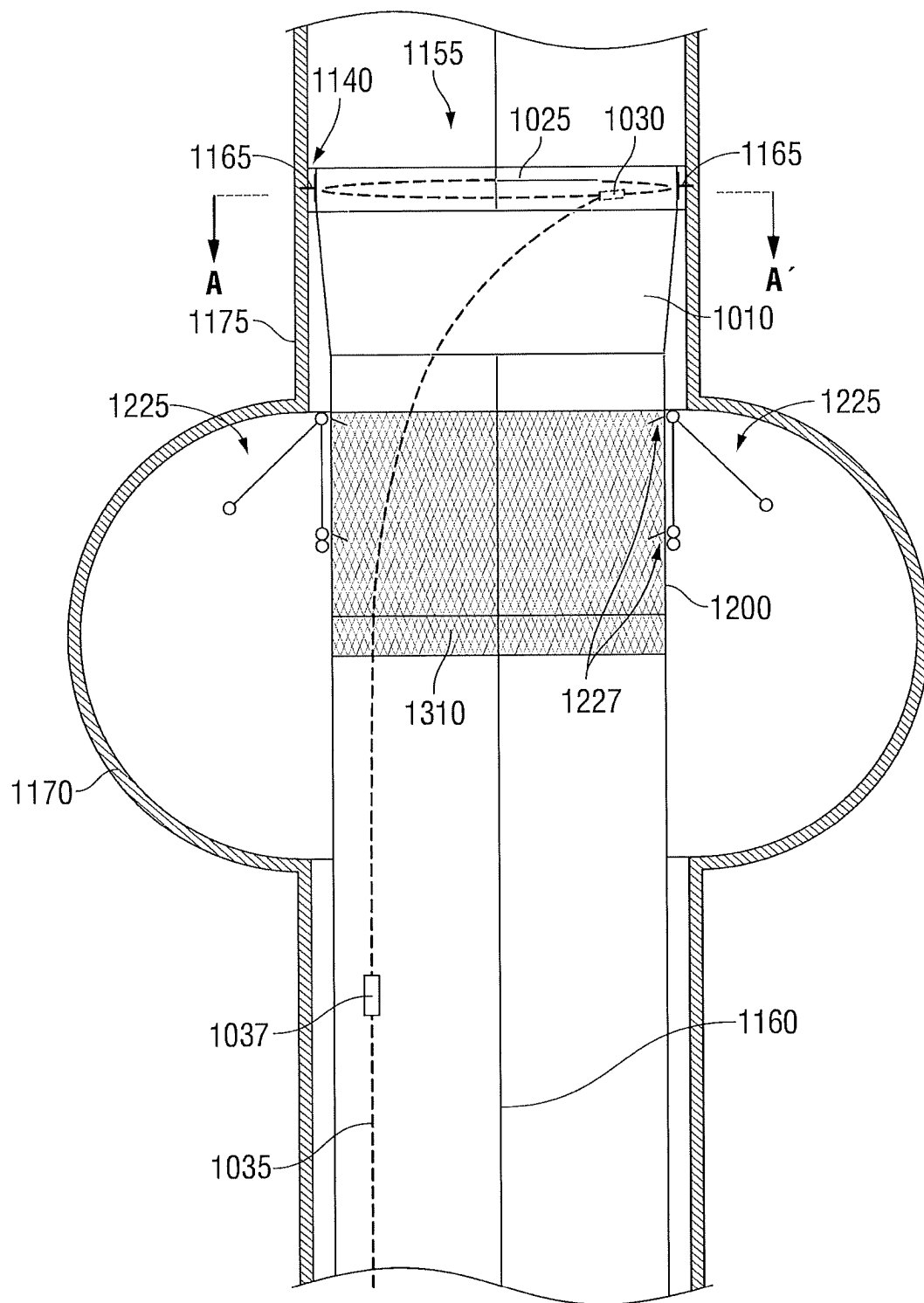


FIG. 12

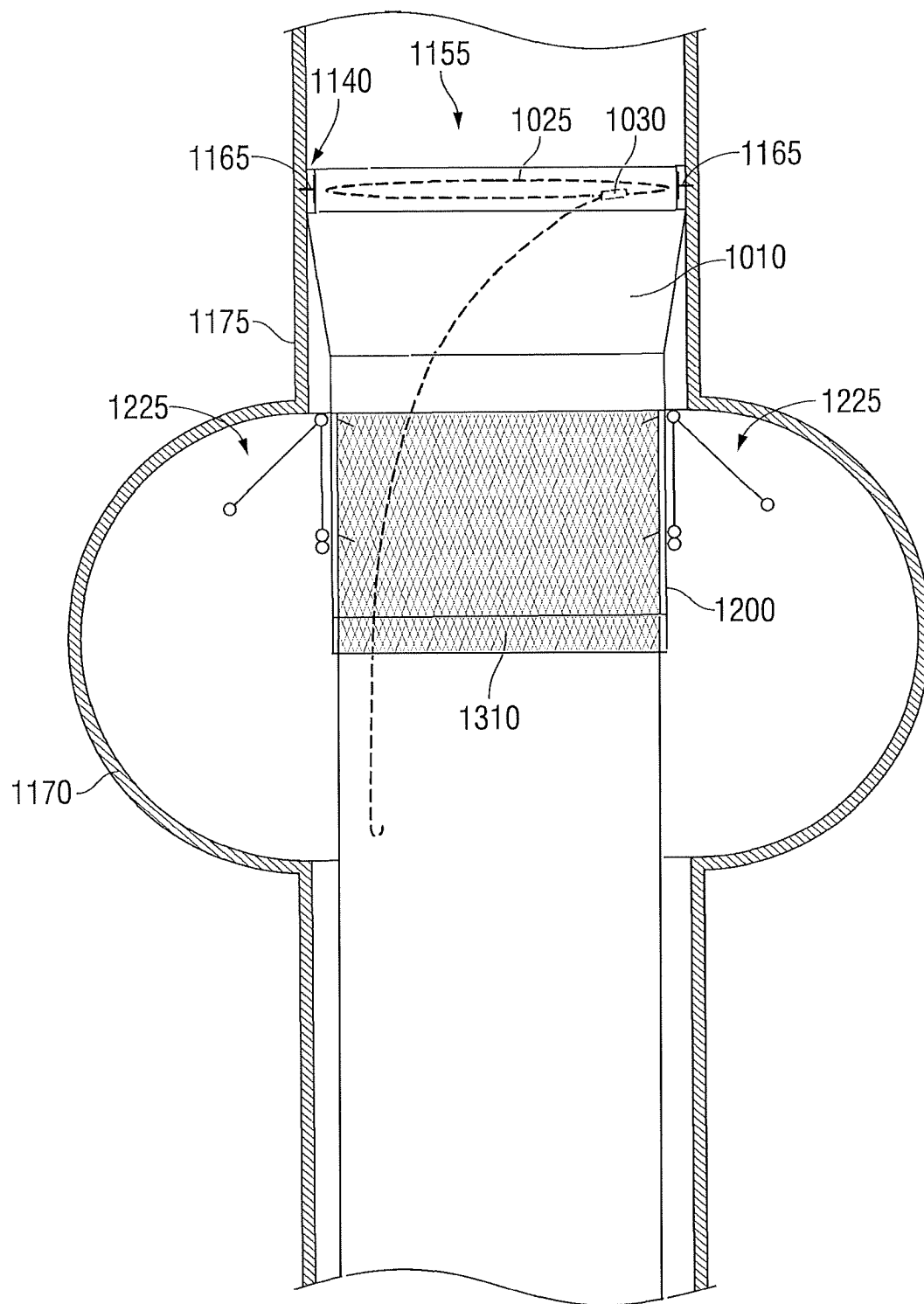


FIG. 13

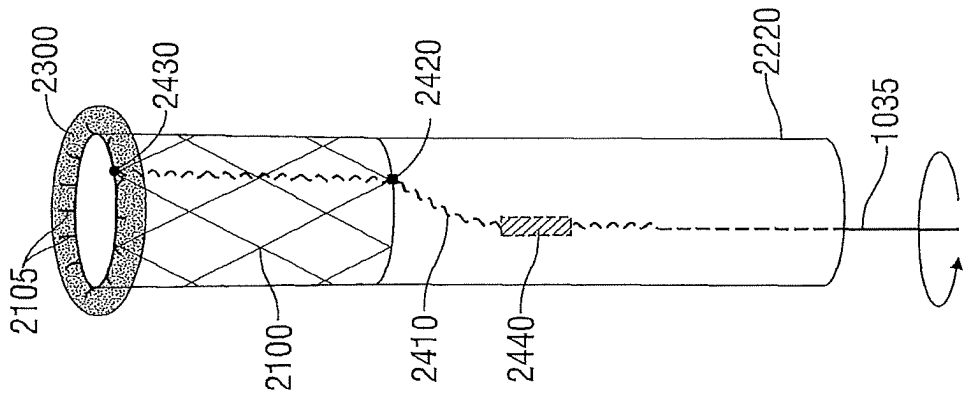


FIG. 14A

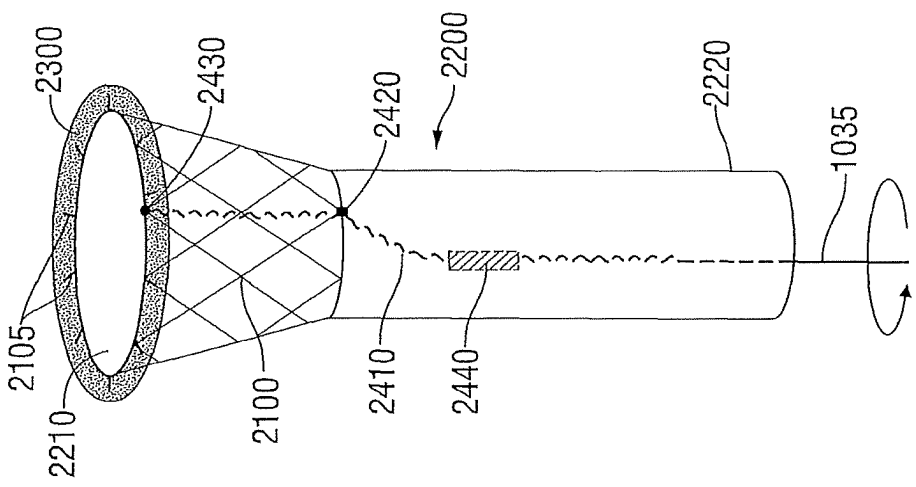


FIG. 14B

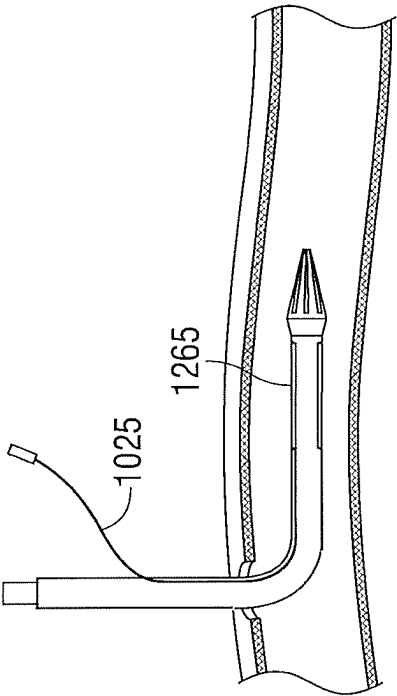


FIG. 15B

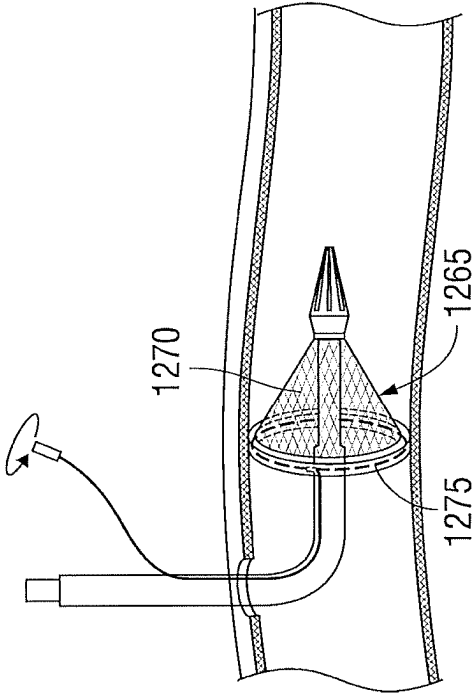


FIG. 15C

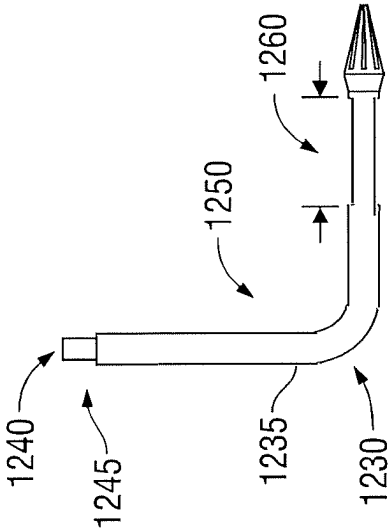


FIG. 15A

SURGICAL IMPLANT DEVICES AND METHODS FOR THEIR MANUFACTURE AND USE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 18/086,059, filed Dec. 21, 2022, which is a continuation of U.S. patent application Ser. No. 16/716,416, filed on Dec. 16, 2019, now U.S. Pat. No. 11,540,911, which is a continuation of U.S. patent application Ser. No. 15/448,417, filed on Mar. 2, 2017, now U.S. Pat. No. 10,507,097, which is a continuation of U.S. patent application Ser. No. 13/339,236, filed Dec. 28, 2011, now U.S. Pat. No. 9,585,743, which claims the benefit of U.S. Provisional Patent Application No. 61/428,114, filed Dec. 29, 2010, all of which are incorporated by reference herein.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not Applicable

FIELD OF THE INVENTION

[0003] The present invention relates to the field of surgical implant devices and methods for their manufacture and use. Among the exemplary embodiments of the present invention are improvements in sealing and retention medical devices particularly applicable to vascular surgery and the treatment of aneurysms or other luminal defects in other anatomic conduits, such as sealing and retention of replacement heart valves.

BACKGROUND OF THE INVENTION

[0004] Medical and surgical implants are placed often in anatomic spaces where it is desirable for the implant to conform to the unique anatomy of the targeted anatomic space and secure a seal therein, preferably without disturbing or distorting the unique anatomy of that targeted anatomic space.

[0005] While the lumens of most hollow anatomic spaces are ideally circular, in fact, the cross-sectional configurations of most anatomic spaces are, at best, ovoid, and may be highly irregular. Such luminal irregularity may be due to anatomic variations and/or to pathologic conditions that may change the shape and topography of the lumen and its associated anatomic wall. Examples of anatomic spaces where such implants may be deployed include, but are not limited to, blood vessels, the heart, other vascular structures, vascular defects (such as thoracic and abdominal aortic aneurysms), the trachea, the oropharynx, the esophagus, the stomach, the duodenum, the ileum, the jejunum, the colon, the rectum, ureters, urethras, fallopian tubes, biliary ducts, pancreatic ducts, or other anatomic structures containing a lumen used for the transport of gases, blood, or other liquids or liquid suspensions within a mammalian body.

[0006] For a patient to be a candidate for existing endograft methods and technologies, to permit an adequate seal, a proximal neck of, ideally, at least 12 mm of normal aorta must exist downstream of the left subclavian artery for thoracic aortic aneurysms or between the origin of the most inferior renal artery and the origin of the aneurysm in the case of abdominal aneurysms. Similarly, ideally, at least 12

mm of normal vessel must exist distal to the distal extent of the aneurysm for an adequate seal to be achieved.

[0007] Migration of existing endografts has also been a significant clinical problem, potentially causing leakage and profusion of aneurysms and/or compromising necessary vascular supplies to arteries such as the carotid, subclavian, renal, or internal iliac vessels. This problem only has been addressed partially by some existing endograft designs, in which barbs or hooks have been incorporated to help retain the endograft at its intended site. However, most existing endograft designs are solely dependent on radial force applied by varying length of stent material to secure a seal against the recipient vessel walls.

[0008] Because of the limitations imposed by existing vascular endograft devices and endovascular techniques, a significant number of abdominal and thoracic aneurysms repaired in the U.S. are still managed through open vascular surgery, instead of the lower morbidity of the endovascular approach.

[0009] Pre-sizing is required currently in all prior art endografts. Such pre-sizing based on CAT-scan measurements is a significant problem. This leads, many times, to mis-sized grafts. In such situations, more grafts segments are required to be placed, can require emergency open surgery, and can lead to an unstable seal and/or migration. Currently there exists no endograft that can be fully repositioned after deployment.

[0010] Thus, a need exists to overcome the problems with the prior art systems, designs, and processes as discussed above.

SUMMARY OF THE INVENTION

[0011] The invention provides surgical implant devices and methods for their manufacture and use that overcome the hereinafore-mentioned disadvantages of the heretofore-known devices and methods of this general type and that provide such features with improvements that increase the ability of such an implant to be precisely positioned and sealed, with better in situ accommodation to the local anatomy of the targeted anatomic site. The invention provides an adjustment tool that can remotely actuate an adjustment member(s) that causes a configuration change of a portion(s) of an implant, which configuration change includes but is not limited to diameter, perimeter, shape, and/or geometry or a combination of these, to create a seal and provide retention of an implant to a specific area of a target vessel or structure.

[0012] One exemplary aspect of the present invention is directed towards novel designs for endovascular implant grafts, and methods for their use for the treatment of aortic aneurysms and other structural vascular defects. An endograft system for placement in an anatomic structure or blood vessel is disclosed in which an endograft implant comprises, for example, a non-elastic tubular implant body with at least an accommodating proximal end. Accommodating, as used herein, is the ability to vary a configuration in one or more ways, which can include elasticity, expansion, contraction, and changes in geometry. Both or either of the proximal and distal ends in an implant according to the present invention further comprise one or more circumferential expandable sealable collars and one or more expandable sealing devices, capable of being expanded upon deployment to achieve the desired seal between the collar and the vessel's inner wall. Exemplary embodiments of such devices can be found in co-pending U.S. patents application

Ser. No. 11/888,009, filed Jul. 31, 2007, and Ser. No. 12/822,291, filed Jun. 24, 2010, which applications have been incorporated herein in their entireties. Further embodiments of endovascular implants according to the present invention may be provided with retractable retention tines or other retention devices allowing an implant to be repositioned before final deployment. In other embodiments, the implant can be repositioned after final deployment. An endograft system according to the present invention further comprises a delivery catheter with an operable tubular sheath capable of housing a folded or compressed endograft implant prior to deployment and capable of retracting or otherwise opening in at least its proximal end to allow implant deployment. The sheath is sized and configured to allow its placement via a peripheral arteriotomy site, and is of appropriate length to allow its advancement into the aortic valve annulus, ascending aorta, aortic arch, and thoracic or abdominal aorta, as required for a specific application.

[0013] While some post-implantation remodeling of the aortic neck proximal to an endovascular graft (endograft) has been reported, existing endograft technology does not allow for the management of this condition without placement of an additional endograft sleeve to cover the remodeled segment.

[0014] Exemplary endografts of the present invention as described herein allow for better accommodation by the implant of the local anatomy, using a self-expandable or compressible gasket for the sealing interface between the endograft collar and the recipient vessel's inner wall. Furthermore, exemplary endografts of the present invention as disclosed herein are provided with a controllably releasable disconnect mechanism that allows remote removal of an adjustment tool and locking of the retained sealable mechanism after satisfactory positioning and sealing of the endograft. In some exemplary embodiments according to the present invention, the controllably releasable disconnect mechanism may be provided in a manner that allows post-implantation re-docking of an adjustment member to permit post-implantation repositioning and/or resealing of an endograft subsequent to its initial deployment.

[0015] In other exemplary applications encompassed by the present invention, improved devices for sealing other medical devices such as vascular cannulae may be provided. The present invention further includes novel designs for vascular cannulae to be used when bi-caval cannulation of the heart is indicated, eliminating the need to perform circumferential caval dissection and further reducing the tissue trauma caused by prior art balloon or other bypass cannulae. While the vascular cannulae of the present invention are inserted and positioned by a surgeon in the standard fashion, the need for circumferential dissection of the cavae and tourniquet placement is obviated. After the vascular cannulae of the present invention are positioned and secured with purse string sutures, the surgeon deploys the adjustable sealing devices of the cannulae by turning an adjustment tool or torque wire. Once the sealing devices are deployed, all of the venous return is diverted. The sealing devices deploy around the distal ends of the cannulae and allow blood to flow through the lumen of the cannulae, but not around the sealing devices. Use of these cannulae minimizes the chance of caval injury by eliminating the need for circumferential dissection. Additionally, the configuration of the adjustable sealing device in relation to the cannula is such that the adjustable sealing device is "flush" with the

cannula so that no acute change in diameter exists along the external surface of the cannula, which serves to avoid tissue trauma during insertion and withdrawal into and out of bodily structures.

[0016] The present invention addresses several major problems presented by existing designs for balloon cannulae. In various exemplary embodiments according to the present invention, the lumens are configured such that a cannula with an adjustable sealing device can be deployed without compromising either the flow within the principle lumen of the cannula or the seal between the cannula and the structure within which the cannula lies. Moreover, a disclosed example of a cannula according to the present invention is provided with a trough within the cannula body at its distal end in which the adjustable sealing device member lies such that, when undeployed during insertion and withdrawal, there is a smooth interface between the external cannula wall and the undeployed sealing device, allowing for smoother, easier, and safer insertion and withdrawal.

[0017] Moreover, existing designs for balloon cannulae are unable to provide a truly symmetrical placement of an inflated balloon around a central lumen of standard diameter. The asymmetry that results with conventional balloon inflation is sufficient to displace the lumen from the true center of the endovascular lumen in which the balloon cannula is placed, resulting in unpredictable and suboptimal flow characteristics therethrough. The altered hemodynamics of such flow with an existing balloon cannula increases the likelihood of intimal vascular injury and clot or plaque embolization. Vascular cannulae of the present invention achieve the surprising result of having the flow characteristics of a non-balloon cannula by maintaining the preferred laminar flow characteristics of a circular main lumen of consistent diameter, positioned and maintained in or near the center of vascular flow by an adjustable sealing device originally provided within a recessed trough in the exterior wall of the cannula, with accessory lumens contained within an externally circular cannular wall. This allows for better seal, less vascular trauma, and easier vascular ingress and egress.

[0018] In addition, vascular cannulae according to the present invention may be provided with retractable stabilizing elements to anchor the inflated balloon within a vessel lumen during use. Such stabilizing elements further make use of the trough within the cannula body, with the stabilizing elements retracting into this trough during insertion and removal, allowing for smooth and trauma-free entry and egress of the cannula.

[0019] Certain aspects of the present invention are directed towards novel designs for sealable endovascular implant grafts, and methods for their use for the treatment of aortic aneurysms and other structural vascular defects or for heart valve replacements. Various embodiments as contemplated within the present invention may include any combination of exemplary elements as disclosed herein or in the co-pending patent applications referenced above.

[0020] In an exemplary embodiment according to the present invention, a sealable vascular endograft system for placement in a vascular defect is provided, comprising an elongated main implant delivery catheter with an external end and an internal end for placement in a blood vessel with internal walls. In such an exemplary embodiment, the main implant delivery catheter further comprises a main implant delivery catheter sheath that may be openable or removable at the internal end and a main implant delivery catheter

lumen containing within a compressed or folded endovascular implant. Further, in such an exemplary embodiment, an endovascular implant comprises a non-elastic tubular implant body with an accommodating proximal end terminating in a proximal sealable circumferential collar that may be expanded by the operator to achieve a fluid-tight seal between the proximal sealable circumferential collar and the internal walls of the blood vessel proximal to the vascular defect. Moreover, in such an exemplary embodiment, an endovascular implant may further comprises a non-elastic tubular implant body with an accommodating distal end terminating in a distal sealable circumferential collar controlled by a distal variable sealing device, which may be expanded by the operator to achieve a fluid-tight seal between the distal sealable circumferential collar and the internal walls of the blood vessel distal to the vascular defect.

[0021] In a further exemplary embodiment according to the present invention, an implant interface is provided for a sealable attachment of an implant to a wall within the lumen of a blood vessel or other anatomic conduit.

[0022] In a yet further exemplary embodiment according to the present invention, an implant gasket interface is provided for a sealable attachment of an implant to a wall within the lumen of a blood vessel or other anatomic conduit, wherein the sealable attachment provides for auto-adjustment of the seal while maintaining wall attachment to accommodate post-implantation wall remodeling.

[0023] Still other exemplary embodiments of endografts and endograft delivery systems according to the present invention serve as universal endograft cuffs, being first placed to offer their advantageous anatomic accommodation capabilities, and then serving as a recipient vessel for other endografts, including conventional endografts.

[0024] Furthermore, exemplary embodiments of endografts and endograft delivery systems according to the present invention may be provided with a mechanism to permit transfer of torque or other energy from a remote operator to an adjustment member comprising a sealable, adjustable circumferential assembly controlled by an adjustment tool, which may be detachable therefrom and may further cause the assembly to lock upon detachment of the tool. In some exemplary embodiments of the present invention, the variable sealing device may be provided with a re-docking element that may be recaptured by subsequent operator interaction, allowing redocking and repositioning and/or resealing of the endograft at a time after its initial deployment.

[0025] Moreover, the various exemplary embodiments of the present invention as disclosed herein may constitute complete endograft systems, or they may be used as components of a universal endograft system as disclosed in co-pending patent applications that may allow the benefits of the present invention to be combined with the ability to receive other endografts.

[0026] Finally, the present invention encompasses sealable devices that may be used in other medical devices such as adjustable vascular cannulas or other medical or surgical devices or implants, such as aortic valves.

[0027] With the foregoing and other objects in view, there is provided, in accordance with the invention, a surgical implant including an implant body and a selectively adjustable assembly attached to the implant body, having adjustable elements, and operable to cause a configuration change

in a portion of the implant body and, thereby, permit implantation of the implant body within an anatomic orifice to effect a seal therein under normal physiological conditions.

[0028] The preceding description is presented only as an exemplary application of the devices and methods according to the present invention.

[0029] Although the invention is illustrated and described herein as embodied in surgical implant devices and methods for their manufacture and use, it is, nevertheless, not intended to be limited to the details shown because various modifications and structural changes may be made therein without departing from the spirit of the invention and within the scope and range of equivalents of the claims. Additionally, well-known elements of exemplary embodiments of the invention will not be described in detail or will be omitted so as not to obscure the relevant details of the invention.

[0030] Additional advantages and other features characteristic of the present invention will be set forth in the detailed description that follows and may be apparent from the detailed description or may be learned by practice of exemplary embodiments of the invention. Still other advantages of the invention may be realized by any of the instrumentalities, methods, or combinations particularly pointed out in the claims.

[0031] Other features that are considered as characteristic for the invention are set forth in the appended claims. As required, detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the invention, which can be embodied in various forms. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one of ordinary skill in the art to variously employ the present invention in virtually any appropriately detailed structure. Further, the terms and phrases used herein are not intended to be limiting; but rather, to provide an understandable description of the invention. While the specification concludes with claims defining the features of the invention that are regarded as novel, it is believed that the invention will be better understood from a consideration of the following description in conjunction with the drawing figures, in which like reference numerals are carried forward.

BRIEF DESCRIPTION OF THE DRAWINGS

[0032] The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views, which are not true to scale, and which, together with the detailed description below, are incorporated in and form part of the specification, serve to illustrate further various embodiments and to explain various principles and advantages all in accordance with the present invention. Advantages of embodiments of the present invention will be apparent from the following detailed description of the exemplary embodiments thereof, which description should be considered in conjunction with the accompanying drawings in which:

[0033] FIG. 1 is a fragmentary, perspective view of an exemplary embodiment of a proximal aspect of a selectively expandable and contractable endograft according to the present invention with the endograft in a relatively expanded form;

[0034] FIG. 2 is a fragmentary, perspective view of the selectively expandable and contractable endograft of FIG. 1 with the endograft in a relatively contracted form;

[0035] FIG. 3 is a fragmentary, perspective view of another exemplary embodiment of a proximal aspect of an endograft according to the present invention further incorporating a lattice structure;

[0036] FIG. 4A is a fragmentary, perspective view of the endograft of FIG. 3 with the endograft in a relatively contracted form;

[0037] FIG. 4B is a fragmentary, perspective view of the endograft of FIG. 3 with the endograft in a partially expanded form;

[0038] FIG. 4C is a fragmentary, perspective view of the endograft of FIG. 3 with the endograft in a fully expanded form;

[0039] FIG. 5A is a fragmentary, partially hidden, perspective view of an exemplary embodiment of a microcylinder locking mechanism with an associated adjustment tool prior to engagement of the microcylinder locking mechanism by the adjustment tool;

[0040] FIG. 5B is a fragmentary, partially hidden, perspective view of the microcylinder locking mechanism and adjustment tool of FIG. 5B with engagement of the microcylinder locking mechanism by the adjustment tool;

[0041] FIG. 5C is a fragmentary, partially hidden, perspective view of an exemplary embodiment of the microcylinder locking mechanism and adjustment tool of FIG. 5B after adjustment and disengagement of the adjustment tool from the microcylinder locking mechanism;

[0042] FIG. 6A is an axial cross-sectional view of the microcylinder and guide bullet along section line A-A of FIG. 5A with tines captures in striations of the microcylinder;

[0043] FIG. 6B is an axial cross-sectional view of the adjustment tool along section line B-B of FIG. 5A;

[0044] FIG. 6C is an axial cross-sectional view of the microcylinder along section line C-C of FIG. 5B;

[0045] FIG. 6D is an axial cross-sectional view of the microcylinder, the guide bullet, and the tool sheath along section line D-D of FIG. 5B without the adjustment member with the tines removed from the microcylinder by the adjustment tool;

[0046] FIG. 6E is an axial cross-sectional view of another exemplary embodiment of a microcylinder locking mechanism and adjustment tool sheath according to the invention where the adjustment tool also has striations having a rectangular cross-sectional shape and has a smooth exterior;

[0047] FIG. 6F is an axial cross-sectional view of yet another exemplary embodiment of a microcylinder locking mechanism according to the invention in which the microcylinder has striations with a triangular cross-sectional shape and with the tines caught in the striations of the microcylinder;

[0048] FIG. 6G is an axial cross-sectional view of the microcylinder locking mechanism of FIG. 6F and an adjustment tool according to the invention in which the tines are removed from the microcylinder by the adjustment tool;

[0049] FIG. 7A is a longitudinal, partial cross-sectional view of an exemplary embodiment of an adjustment control locking mechanism according to the present invention with a controllable catch mechanism disengaged;

[0050] FIG. 7B is a longitudinal, partial cross-sectional view of the adjustment control locking mechanism of FIG. 7A with the controllable catch mechanism engaged.

[0051] FIG. 8A is a fragmentary, partially hidden, perspective view of an exemplary embodiment of a microcylinder locking mechanism according to the invention with internal locking tines of unequal length and with an associated adjustment tool sheath prior to engagement of the microcylinder locking mechanism by the adjustment tool sheath;

[0052] FIG. 8B is a fragmentary, partially hidden, perspective view of the microcylinder locking mechanism and adjustment tool sheath of FIG. 7A with engagement of the microcylinder locking mechanism by the adjustment tool sheath;

[0053] FIG. 8C is a fragmentary, partially hidden, perspective view of the microcylinder locking mechanism and adjustment tool sheath of FIG. 7B after adjustment and disengagement of the microcylinder locking mechanism with the adjustment tool sheath.

[0054] FIG. 9A is an axial cross-sectional view of retention tines sheathed by an expanded compressible foam gasket in an exemplary endograft according to the present invention with the tines in a non-extended state;

[0055] FIG. 9B is a fragmentary, perspective view of the retention tines of FIG. 9A exposed and deployed through a compressible foam gasket by an expanded sealable collar in an exemplary endograft according to the present invention;

[0056] FIG. 10A is a fragmentary, axial cross-sectional view of an exemplary endovascular interface cuff according to the present invention, in which the interface cuff has been positioned over an endovascular guidewire to a desired recipient site in the aorta proximal to an aortic aneurysm sac but has not been expanded therein;

[0057] FIG. 10B is a fragmentary, transverse cross-sectional view of the interface cuff of FIG. 10A;

[0058] FIG. 11A is a fragmentary, axial cross-sectional view of the interface cuff of FIG. 10A, with expansion of the endovascular interface cuff in the aorta to achieve a seal and with retention tine engagement of the aortic wall in the desired recipient site proximal to the aortic aneurysm sac at the level of A-A';

[0059] FIG. 11B is a fragmentary, transverse cross-sectional view of the interface cuff of FIG. 11A;

[0060] FIG. 12 is a fragmentary, axial cross-sectional view of the interface cuff of FIG. 10A with delivery of an endograft secured within the rigid cuff of the interface cuff;

[0061] FIG. 13 is a fragmentary, axial cross-sectional view of the interface cuff of FIG. 12 with the guidewire removed and with the adjustment tool detached and removed;

[0062] FIG. 14A is a fragmentary, perspective view of an exemplary embodiment of an actively controllable endograft according to the present invention in which a latticework external to the lumen of an endograft can be radially displaced by controlled rotation of an adjustment member, the lattice structure being in a contracted state;

[0063] FIG. 14B is a fragmentary, perspective view of the actively controllable endograft of FIG. 14A in which the lattice structure is in an expanded state;

[0064] FIG. 15A is a side perspective view of an exemplary embodiment of an adjustable vascular cannula according to the present invention;

[0065] FIG. 15B is a side perspective and partially hidden view of the adjustable vascular cannula of FIG. 15A within

a recipient blood vessel with an adjustable seal device in a non-deployed, contracted position; and

[0066] FIG. 15C is a side perspective and partially hidden view of the adjustable vascular cannula of FIG. 15B with the adjustable seal device in a deployed, expanded position.

DETAILED DESCRIPTION OF THE INVENTION

[0067] As required, detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the invention, which can be embodied in various forms. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the present invention in virtually any appropriately detailed structure. Further, the terms and phrases used herein are not intended to be limiting; but rather, to provide an understandable description of the invention. While the specification concludes with claims defining the features of the invention that are regarded as novel, it is believed that the invention will be better understood from a consideration of the following description in conjunction with the drawing figures, in which like reference numerals are carried forward.

[0068] Alternate embodiments may be devised without departing from the spirit or the scope of the invention. Additionally, well-known elements of exemplary embodiments of the invention will not be described in detail or will be omitted so as not to obscure the relevant details of the invention.

[0069] Before the present invention is disclosed and described, it is to be understood that the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. The terms “a” or “an”, as used herein, are defined as one or more than one. The term “plurality,” as used herein, is defined as two or more than two. The term “another,” as used herein, is defined as at least a second or more. The terms “including” and/or “having,” as used herein, are defined as comprising (i.e., open language). The term “coupled,” as used herein, is defined as connected, although not necessarily directly, and not necessarily mechanically.

[0070] Relational terms such as first and second, top and bottom, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” or any other variation thereof are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.

[0071] As used herein, the term “about” or “approximately” applies to all numeric values, whether or not explicitly indicated. These terms generally refer to a range of numbers that one of skill in the art would consider equivalent to the recited values (i.e., having the same function or

result). In many instances these terms may include numbers that are rounded to the nearest significant figure.

[0072] Herein various embodiments of the present invention are described. In many of the different embodiments, features are similar. Therefore, to avoid redundancy, repetitive description of these similar features may not be made in some circumstances. It shall be understood, however, that description of a first-appearing feature applies to the later described similar feature and each respective description, therefore, is to be incorporated therein without such repetition.

[0073] Described now are exemplary embodiments of the present invention. Referring now to the figures of the drawings in detail and, first, particularly to FIG. 1 thereof, there is shown a perspective view of an exemplary embodiment of the proximal aspect of a sealable endograft system 1000 according to the present invention, in which the endograft is in a relatively expanded form. FIG. 2 is a perspective view of the embodiment of the proximal aspect of a sealable endograft system 1000 according to the present invention of FIG. 1, showing the endograft in a relatively contracted form. This exemplary endograft system 1000 has the ability to be selectively expanded and contracted to a diameter selected by the implanting physician. In general, the endograft system 1000 has, along its intermediate extent and, possibly, also at its distal portion (at the downstream end of the prosthesis), a relatively constant diameter portion. At its proximal portion (at the upstream end of the prosthesis), the endograft system 1000 is able to impart a configuration change to selectively adjustable portion of the implant. Features of the inventive controllable endograft system 1000 are described in further detail in U.S. patents application Ser. No. 11/888,009, filed Jul. 31, 2007, and Ser. No. 12/822,291, filed Jun. 24, 2010, which have been incorporated herein and detail of which is not replicated herein for the sake of brevity.

[0074] The exemplary sealable endograft system 1000 shown in FIGS. 1 and 2 comprises a hollow tubular endograft body 1005 having an accommodating proximal cuff 1010 and an intermediate, substantially rigid, tubular member 1015. The distal end of such an endograft (not shown in FIGS. 1 and 2) may be any or all of accommodating, elastic, rigid, stent-laden, or even replicate the proximal end, depending upon the various exemplary embodiments according to the present invention. A selectively adjustable circumferential assembly 1020 is disposed at the proximal cuff 1010. Contained in one exemplary embodiment of the circumferential assembly 1020 is a circumferential channel enclosing an adjustment member 1025 (indicated only diagrammatically with a solid line). The adjustment member 1025 causes the expansion/contraction of the accommodating proximal cuff 1010 by looping around the perimeter and by being lengthened or shortened, respectively. The adjustment member 1025, for example, interacts with a control device 1030 that is operable to cause an increase or decrease in the circumference of the circumferential loop 1025 by the application of rotational torque to the distal aspect of an adjustment tool 1035 emerging from the control device 1030. The adjustment member 1025 can be integral with the adjustment tool 1035 in an exemplary embodiment of the circumferential assembly 1020, or can be removable as shown, for example, in FIG. 10A.

[0075] Such an adjustment member 1025 may take many forms in the present invention. In one exemplary embodi-

ment according to the present invention, the adjustment member **1025** is a micro-threaded cable that is fixed at one end to the control device **1030**, which is in the form of a microcylinder, and the adjustment tool **1035** threads through a threaded aspect of the microcylinder **1030** in order to effect a change in the circumference of the proximal cuff **1010**. A forwardly imposed torque on the adjustment tool **1035** cause expansion of the adjustment tool **1035**. Expansion of the adjustment member **1025** in its circumferential extent has the effect of expanding the proximal aspect of the sealable endograft system **1000** to allow for precise sealing of the sealable endograft system **1000** within a recipient blood vessel such as the aorta (not shown in FIG. 1 or 2). Conversely, reverse torque on the adjustment tool **1035** has the effect of decreasing the circumference of the circumferential loop of the adjustment member **1025** and, thus, contracting the proximal aspect of the sealable endograft system **1000**, allowing for re-positioning as needed. In FIGS. 1 and 2, the adjustment tool **1035** may extend distally through the lumen of the sealable endograft system **1000**. Alternatively, the adjustment tool **1035** may extend distally through a separate lumen provided in the sealable endograft system **1000** (not shown in FIG. 1 or 2).

[0076] FIGS. 3 and 4A to 4C are perspective views of yet another exemplary embodiment of a proximal aspect of a sealable endograft system **1000** according to the present invention that further incorporates a stent or lattice structure **1041** (which, in another embodiment, can be a compressible foam gasket). The lattice structure **1041** is provided with a lattice interruption **1045** to allow for variations in the circumference of the proximal aspect of the endograft. This lattice interruption **1045** may take the form of a V-shape as shown in FIGS. 4B and 4C or may be otherwise configured. As in FIGS. 1 and 2, the sealable endograft system **1000** of FIG. 3 also has an accommodating proximal cuff **1010** which encloses the terminal lattice structure **1040** as shown and also encloses an adjustment member **1025** that loops through a control device **1030** that is provided to allow increase or decrease in the circumference of the, e.g., circumferential loop of the adjustment member **1025** by the application of rotational torque to the distal aspect of the adjustment tool **1035** emerging from the control device **1030**. The progression of FIGS. 4A to 4C shows the endograft in a relatively contracted form in FIG. 4A, in a partially expanded form in FIG. 4B, and in a fully expanded form in FIG. 4C. As the lattice interruption **1045** is closed in FIGS. 3 and 4A, it can be seen only in FIGS. 4B and 4C. One exemplary configuration for the lattice interruption **1045** can be a woven material that is stretched in the expanded state and attached to the lattice **1041** and, when allowed to reduce, the woven material resist buckling. This configuration allows the diameter to increase beyond the maximum diameter that the graft will allow with the stent alone.

[0077] FIG. 5A shows an exemplary embodiment of the control device **1030** in the form of a microcylinder locking mechanism **1050**. This locking mechanism **1050** is changed from a locked state to an unlocked state by an adjustment tool **1060**, which comprises a tool sheath **1062** having a keyed collar portion **1065**. The adjustment tool **1060** is fixed, in both the longitudinal and radial extents, to the remote adjustment tool **1035**. The progression of FIGS. 5A to 5C show how the locking mechanism **1050** is changed from the locked state (in which adjustment of the adjustment member **1025** is prohibited) to the unlocked state (in which adjust-

ment of the adjustment member **1025** is permitted), and, then, back to the locked state.

[0078] Before explaining the change between states, the configuration of an exemplary embodiment of the locking mechanism **1050** is described further. The exterior of the locking mechanism **1050** is comprised of a microcylinder **1052** having a set of circumferentially spaced-apart, interior striations **1055**. The locking mechanism **1050** is longitudinally and rotationally fixed to the proximal cuff **1010**. A guide bullet **1070** is received within the hollow, internally striated microcylinder **1052**. The guide bullet **1070** has a longitudinal threaded bore that received therein (in a threaded manner) the adjustment member **1025**. The adjustment member **1025** completely traverses the bore of the guide bullet **1070** and terminates distally of the guide bullet **1070** in a keyed block **1075** that is rotationally fixed to the adjustment member **1025**. The guide bullet **1070** has at least two opposing, flexible tines **1072** that extend radially outward, in a natural state that, together, has a diameter greater than the internal diameter of the locking microcylinder **1052** (the tines can, as well, be spring loaded outwardly). The tines **1072** have a terminal portion that is shaped to fit within a corresponding shaped of each striation **1055** within the microcylinder **1052**. As such, when the tines **1072** are compressed and the guide bullet **1070** is placed within the microcylinder with the adjustment member **1025** threaded therewithin, the tines **1072** press outwardly against the internal surface of the microcylinder **1052** and, when appropriately rotated therein, the tines **1072** each lock within a respective opposing one of the striations **1055**. In such a state, the tines **1072** both form-fittingly and force-fittingly lock within inner striations **1055** when unconstrained. If, for example, there were three tines **1072** separated by 120 degrees each, then the tines **1072** would each lock within a respective one of the striations **1055** that are, also, 120 degrees apart along the interior surface of the microcylinder **1052**. The frictional force of the tines **1072** against the inside surface of the microcylinder **1052** is sufficiently strong to prevent longitudinal movement of the guide bullet **1070**, even if the keyed block **1075** is rotated unless the tines **1072** are removed from their locked position against the interior surface of the microcylinder. In such a configuration, the microcylinder **1052** and the guide bullet **1070** prevent rotation of the adjustment member **1025** without, not only a particular external force applied thereto, but also a removal of the tines **1072** from the interior surface of the microcylinder **1052**.

[0079] Rotation of the adjustment member **1025**, therefore, is carried out with the adjustment tool **1060**. The adjustment tool **1060** provides both the ability to rotate the keyed block **1075** but also the ability to separate the tines **1072** from the interior surface of the microcylinder **1052**. To carry out these functions, the tool sheath **1062** has a sufficient cylindrical length to slide between the tines **1072** and the interior surface of the microcylinder **1052** anywhere the tines **1072** are contacting the interior surface. As such, the longitudinal length of the tool sheath **1062** can be, but does not necessarily have to be, as long as the microcylinder **1052**. FIG. 5A shows the microcylinder **1052** with the guide bullet **1070** in a locked position, prior to interface by the remote adjustment tool **1060**. When the adjustment tool **1060** is slid into the microcylinder **1052**, as shown in the progression of FIGS. 5A to 5B, the smooth interior surface of the tool sheath **1062** first slides along the outer surface of

the tines and, then, along and past the distal ends of the tines **1072**, at which time the tines **1072** no longer contact the interior surface of the microcylinder **1052**. The orientation of the microcylinder locking mechanism **1050** and the adjustment tool **1060** in FIG. **5B** now allows for repositioning of the adjustment member **1025** and relocation of the guide bullet **1070** within the microcylinder **1052**.

[0080] The keyed collar portion **1065** has a distal taper **1067** that reduces the outer diameter of the tool sheath **1062** inwards to such an extent that it acts as a funnel to direct the keyed block **1075** directly into the radial center of the keyed collar portion **1065**. At the proximal-most end of the collar portion **1065** is an internal key **1069** having an internal circumferential shape corresponding to an external circumferential shape of the keyed block **1075**. As such, when the adjustment tool **1060** is inserted into the microcylinder **1052** and releases the tines **1072** from the interior surface thereof, the tool sheath **1062** can pass the tines **1072** (wherever they may be inside the microcylinder **1052**) sufficiently far to permit the keyed block **1075** to slide along the interior distal taper **1067** and press against the internal bore of the key **1069**. With slight rotation either way of the adjustment tool **1060** (by rotation of the adjustment tool **1035**), the keyed block **1075** will fall into the internal bore of the key **1069** in a form-fit, thereby enabling rotation of the adjustment member **1025** (via keyed block **1075**) in a corresponding manner to any rotation of the adjustment tool **1035** by a user.

[0081] The locking mechanism **1050** is longitudinally and rotationally fixed to the circumferential assembly **1020** such that rotation of the locking mechanism **1050** in a first direction causes a contraction of the circumferential assembly **1020** and rotation of the locking mechanism **1050** in the opposition direction causes an expansion of the circumferential assembly **1020**. As can be seen in FIGS. **5B** and **5C**, the keyed block **1075** is rotated to cause the guide bullet **1070** to advance towards the keyed block **1075**. FIG. **5C** shows the microcylinder locking mechanism **1050** with the adjustment tool **1060** after adjustment and disengagement of the microcylinder locking mechanism **1050** by the adjustment tool **1060** with a fixed repositioning of the guide bullet **1070** and a distal lengthening of the adjustment member **1025** with respect to the microcylinder **1052**. As the final position of the keyed block **1075** is further away from the microcylinder **1052**, and because the microcylinder **1052** is fixed to the control device **1030** of the circumferential assembly **1020**, this exemplary movement of the adjustment member **1025** indicates that the circumferential assembly **1020** has reduced in diameter.

[0082] Various alternative embodiments of this locking mechanism are envisioned where a number of the individual parts are fixed or moving with respect to other ones of the parts of the circumferential assembly **1020**, the control device **1030**, the locking mechanism **1050**, and/or the adjustment tool **1060**. In one alternative embodiment of the microcylinder locking mechanism **1050**, the collar portion **1065** of the remote adjustment tool **1060** can contain inner striations (similar to or different from the striations **1055** of the microcylinder **1052**) that allow it to capture and turn the guide bullet **1070** through removable fixation of the tines **1072** therein (see FIG. **6E**). In such a configuration, the guide bullet **1070** can be fixed rotationally to the adjustment member **1025**.

[0083] The inner striations **1055** of the microcylinder **1052** may be grooves, threads, detents, slots, or other surface

features sufficient to allow capture of the tines **1072** upon their release as shown in further detail, for example, in the cross-sections of FIGS. **6A** to **6G**. FIG. **6A** is a cross-section along section line A-A of the microcylinder **1052** and guide bullet **1070** of FIG. **5A**, in which the tines **1072** having an exemplary triangular cross-sectional shape are caught within two striations **1055** having an exemplary rectangular cross-sectional shape. FIG. **6B** is a cross-section along section line B-B of the tool sheath **1062** of FIG. **5A** and illustrates the relatively smooth outer surface of the tool sheath **1062**. FIG. **6C** is a cross-section along section line C-C of the microcylinder **1052** of FIG. **5B** without the adjustment member **1025** depicted. FIG. **6D** is a cross-section along section line D-D of the microcylinder **1052**, the guide bullet **1070**, and the tool sheath **1062** of FIG. **5B**, in which the tool sheath **1062** captures the guide bullet **1070** and collapses the tines **1072**, thereby removing the tines **1072** from the striations **1055** of the microcylinder **1052**.

[0084] FIG. **6E** shows a cross-sectional view of a variation of another exemplary embodiment of the locking mechanism **1050'** with the adjustment tool sheath **1062'** also having striations **1055'** with an exemplary rectangular cross-sectional shape. The tines **1072** are illustrated as expanded within two opposing striations **1055'** of the tool sheath **1062'**. As the tool sheath **1062'** has a smooth exterior, the tool sheath **1062'** can rotate without friction within the microcylinder **1052'**.

[0085] FIGS. **6F** and **6G** show cross-sectional views of yet another variation of an exemplary embodiment of the microcylinder locking mechanism **1050"** and adjustment tool **1060"**. The locking mechanism **1050"** has a microcylinder **1052"** with striations **1055"** having an exemplary triangular cross-sectional shape. The adjustment tool sheath **1062"** has a smooth exterior and interior to slide within the microcylinder **1052"** and to slidably capture the tines **1072"**, respectively. The tines **1072"** are illustrated as expanded within two opposing triangular striations **1055"** of the microcylinder **1052"** in FIG. **6F** and are captured within the tool sheath **1062"** in FIG. **6G**.

[0086] FIGS. **7A** and **7B** show longitudinal cross-sectional details of one exemplary embodiment of a locking mechanism **1110** for the adjustment tool **1035** according to the present invention. FIG. **7A** shows a locking mechanism **1110** comprising a controllable catch **1115** in a disengaged state. FIG. **7B** shows the locking mechanism **1110** with the controllable catch mechanism **1115** engaged. Once the adjustment member catch **1120** is within the target range **1117** of the locking mechanism, the user can engage a non-illustrated catch deployment device to capture the adjustment member catch **1120**.

[0087] FIGS. **8A** to **8C** show details of still another embodiment of a microcylinder locking mechanism **1150** according to the present invention, in which internal locking tines **1152**, **1154** of unequal length are employed to prevent back rotation from torque buildup upon detachment of the remote adjustment tool **1060**. FIG. **8A** shows the locking mechanism **1150** comprised of a microcylinder **1151** and a guide bullet **1153** with internal locking tines **1152**, **1154** of unequal length and an associated adjustment tool **1160** having a tool sheath **1164** prior to engagement of the microcylinder locking mechanism **1150** by the tool sheath **1164**. FIG. **8B** shows the tool sheath **1164** of FIG. **8A** engaged with the microcylinder locking mechanism **1150** to deflect the tines **1152**, **1154** away from the interior surface

of the microcylinder **1151**. FIG. **8C** shows the microcylinder locking mechanism **1150** in a locking position different from FIG. **8A** after adjustment has occurred and the tool sheath **1164** has been disengaged from the microcylinder **1151**.

[0088] FIGS. **9A** and **9B** show two aspects of details of sheathable retention tines **1130** and a compressible foam sealing gasket **1140** for the proximal terminal aspect of some exemplary embodiments of endografts according to the present invention. FIG. **9A** is an axial cross section showing sheathable retention tines **1130** sheathed by an expanded compressible foam gasket **1040** in an exemplary proximal aspect of a sealable endograft system **1000** according to the present invention. FIG. **9B** is a perspective view showing sheathable retention tines **1130** exposed and deployed through the compressible foam sealing gasket **1140** disposed at an expanded proximal cuff **1010** in an exemplary endograft according to the present invention. In some exemplary embodiments of the present invention, the direct pressure of the adjustment member **1025** on the footplate **1145** of the tines may be used to extend the sheathable tines **1130** through the compressible foam gasket **1040** and into the wall of a recipient blood vessel. In yet other exemplary embodiments of the present invention, direct pressure of the adjustment member **1025** may exert force on non-illustrated footplate bands that may be attached to or adjacent the footplates **1145** of the tines **1130** and may be used to extend the sheathable tines **1130** through the compressible foam gasket **1040** and into the wall of a recipient blood vessel. Such footplate bands may, themselves, be the base of the sheathable tines **1130** in certain exemplary embodiments of the present invention. Not shown in FIGS. **9A** and **9B**, the adjustment member **1025** may course through eyelets, other brackets or may otherwise be moveably connected to the footplates **1145** to maintain equal pressure and desired orientation upon expansion of the adjustment member loop.

[0089] In the various embodiments of sealable endograft systems according to the present invention, the distal attachment of the endograft to the aortic wall distal to the aneurysm sac may be accomplished in a conventional manner using an expandable lattice component at the distal cuffs, or variations on the adjustable, sealable mechanism disclosed herein may be employed to secure distal seals. The distal seals are subject to lower pressure demands, and the anatomic constraints of sufficient aortic neck distally are generally less problematic than for the proximal seal.

[0090] FIGS. **10** to **13** provide anatomic views of another exemplary embodiment of an endograft implant according to the present invention in which the implant is a universal proximal cuff endovascular implant for treatment of an abdominal aortic aneurysm. Endografts with the features shown in the various embodiments of the present invention have unique abilities to accommodate to anatomic variations that would preclude or compromise use of conventional endograft systems. The universal proximal cuff implants of the present invention allow an operator to make use of their ability to securely seal and attach in anatomic sites where conventional endografts cannot be securely placed, and then allow a conventional endograft to securely dock with the universal proximal cuff endovascular implants distally.

[0091] Universal proximal cuff endovascular implants of the present invention may be provided with any of the elements disclosed in the present and the incorporated co-pending applications referenced herein. Such elements include, but are not limited to, attachment of radio-opaque

monitoring clip assemblies on the outer surfaces of endografts to allow post-implantation monitoring of slippage or endoleak formation by plain radiographs, steerable delivery systems to permit delivery and seal of an endograft in an anatomically angulated or irregular site, and/or auto-accommodation for post-implantation aortic remodeling.

[0092] FIG. **10A** is an axial cross-sectional view of an exemplary endovascular universal interface cuff **1155** of the present invention to be implanted into an aorta having an aneurysm sac **1170** and an aortic wall **1175**. The universal endovascular interface cuff **1155** has been positioned over an endovascular guidewire **1160** to a desired recipient site A-A' proximal to the aortic aneurysm sac **1170**. The endovascular universal interface cuff **1155** further comprises an accommodating proximal cuff **1010** and a rigid distal cuff **1200**. FIG. **10B** provides a transverse cross-sectional view of the exemplary endovascular interface cuff **1155** of FIG. **10A** at the level of A-A' in FIG. **10A**. In FIGS. **10A** and **10B**, the compressible foam gasket **1140** is uncompressed and, therefore, covers the retention tines **1165**.

[0093] In the exemplary embodiment shown in FIG. **10B**, the adjustment member **1025** courses in a circumferential loop through eyelets **1180** attached to a series of compression footplates **1185**. The compression footplates **1185**, among other functions, serve to maintain an orientation of the expanding circumferential loop **1035** in a plane transverse to the aortic lumen **1190**, and present a broader pressure contact with the underlying aortic wall **1175** when the circumferential assembly is expanded. The compression footplates **1185** may abut, be attached to, or be contiguous with the retention tines **1165**, which are displaced through the compressed compressible foam gasket **1140** and allowed to enter the aortic wall **1175** for overall device stabilization and retention. While four retention tines **1165** and footplates **1185** are shown, this embodiment is merely exemplary and can be any number.

[0094] FIG. **11A** shows the same axial cross-sectional view of the endovascular universal interface cuff **1155** of FIG. **10A** but after the universal endovascular interface cuff **1155** has expanded to achieve a seal in the aortic wall **1175**. Due to the expansion of the cuff, the foam gasket **1140** becomes compressed, allowing the retention tines **1165** to protrude radially outward to engage the aortic wall **1175** in the desired recipient site A-A' proximal to the aortic aneurysm sac **1170**. In the exemplary embodiment shown in FIG. **11B**, the adjustment member **1025** has expanded to move the eyelets **1180** attached to the footplates **1185** outwards. As is evident, the interior lumen of the circumferential assembly **1020** shown in FIG. **11B** has increased substantially as compared to the state shown in FIG. **10B**. In FIG. **11B**, the compression of the foam gasket **1140** and the engagement of the aortic wall **1175** by the retention tines **1165** creates a firm seal between the universal endovascular interface cuff **1155** and the aortic wall **1175**.

[0095] FIG. **12** shows the same axial cross-sectional axial of the universal endovascular interface cuff **1155** of the present invention as in FIGS. **10A** and **11A** but with delivery of a conventional endograft **1300** into the aortic wall **1175**, which endograft **1300** has been secured within the rigid distal cuff **1200** of the universal endovascular interface cuff **1155**. The endograft **1300** can include an expandable lattice **1310**. FIG. **13** shows the same cross-sectional axial view of an exemplary universal endovascular interface cuff **1155** of the present invention as FIG. **12** but after removal of the

endovascular guidewire **1160** and detachment and removal of the adjustment member **1025**. Such removal and detachment can be carried out by a release mechanism **1037**. The distal attachment of the conventional endograft is not shown in FIGS. **12** and **13**, but can be accomplished in the usual manner for conventional endograft implantation sufficient to prevent backfill of the aneurysm sac **1170** from the distal aorta or the iliac vessels.

[0096] As shown in FIGS. **10A**, **11A**, **12**, and **13**, the rigid distal cuff **1200** includes, at its exterior, exemplary radio-opaque monitoring clip assemblies **1225** to allow post-implantation monitoring of slippage or endoleak formation and/or auto-accommodation for post-implantation aortic remodeling. Likewise, the rigid distal cuff **1200** can be provided with interior graft retention tines **1227** that add to securing, without leaks, the endograft **1300** to the interior of the rigid distal cuff **1200**.

[0097] The tubular endograft body **1005**, the proximal cuff **1010**, the rigid distal cuffs **1200**, and the endograft body **1300** as described herein may be constructed of solid, woven, non-woven, or mesh materials such as, but not limited to, natural or synthetic rubbers, nylon, GORE-TEX®, elastomers, polyisoprenes, polyphosphazenes, polyurethanes, vinyl plastisols, acrylic polyesters, polyvinylpyrrolidone-polyurethane interpolymers, butadiene rubbers, styrene-butadiene rubbers, rubber lattices, DACRON®, PTFE, malleable metals, other biologically compatible materials or a combination of such biologically compatible materials in a molded, woven, or non-woven configuration, coated, non-coated, and other polymers or materials with suitable resilience and pliability qualities. In certain exemplary embodiments according to the present invention, it is desirable for the non-elastic tubular member **1015** and corresponding structures to be pliable to allow for folding or compressibility without allowing elasticity. In certain exemplary embodiments according to the present invention, it is desirable for the accommodating proximal cuff **1010** and corresponding structures to have plasticity and be compressible or foldable. In any given exemplary embodiment, the non-elastic tubular implant body **1015**, the endograft body **1300**, the accommodating proximal cuff **1010**, and corresponding structures may be constructed of the same material of varying elasticity, or these structures may be constructed of different, but compatible materials.

[0098] The adjustment members **1025**, the retention tines **1130**, **1165**, and the microcylinders **1030** and other mechanical components as disclosed herein and in all other embodiments of the present invention may be fabricated of any suitably strong biocompatible material, including, but not limited to titanium, stainless steel, cobalt chromium alloys, other metals, other metal alloys, nitinol, plastics, or ceramics. Similarly, the adjustment members **1025**, the retention tines **1130**, **1165**, and the microcylinders **1030** and other mechanical components may be milled, laser cut, lathed, molded, or extruded.

[0099] The compressible foam gaskets **1140** as disclosed herein may be any biocompatible foam material of either an open or closed cell structure with sufficient compressibility and resilience to allow rapid recovery in a non-compressed state. In various exemplary embodiments according to the present invention, such foam materials may be viscoelastic foam with a compressible cellular material that has both elastic (spring-like) and viscous (time-dependent) proper-

ties. Viscoelastic foam differs from regular foam by having time-dependent behaviors such as creep, stress relaxation, and hysteresis.

[0100] FIGS. **14A** and **14B** show an alternate exemplary embodiment of a sealable endograft system **2000** according to the present invention in two different states. In the view of FIG. **14A**, a hinged lattice structure **2100** is attached to an internal or external surface of at least the proximal portion **2210** of an endograft body **2200** (the “lattice” in these figures is only diagrammatic and is not intended to imply that the only possible number of rings of lattice is greater than one). Either the lattice structure **2100** or the endograft body **2200** can be provided with radially displaced retention tines **2105** that, in a non-distended state of the proximal portion **2210**, can be covered within a compressible foam gasket **2300**. In the embodiment shown in FIG. **14A**, the distal portion **2220** of the endograft body **2200** comprises a non-distensible material and the proximal portion **2210** of the endograft body **2200** is an accommodating cuff comprising a distensible material forming the proximally terminal aspect of the sealable endograft system **2000** and enclosing the terminal hinged lattice structure **2100** therewithin.

[0101] A control system **2400** or jack screw shown in FIGS. **14A** and **14B** is provided to expand and contract the lattice structure **2100**. In particular, a torque wire **2410** can be fixed at two points **2420**, **2430** longitudinally separate from one another on the lattice structure **2100**. This torque wire **2410** has exterior threads that correspond to threaded bores of one of the two points **2420**, **2430**. Accordingly, when the torque wire **2410** is rotated, the two points **2420**, **2430** of the lattice either approach one another (to expand the proximal portion **2210**) or retreat from one another (to contract the proximal portion **2210**) this imparts motion to all contiguously interconnected lattice elements. It is preferred to have the proximal end point **2430** be bored for rotation but fixed longitudinally. In this case, a smooth-bored collar **2440** is fixed to the wall of the graft **2200**, for example, on an interior surface distal of the lattice structure **2100**. When the adjustment tool **1035** is rotated, the torque wire **2410** correspondingly rotates to expand or contract the proximal portion **2210** of the endograft **2200**. In this manner, in comparison to self-expanding prior art stent structures (e.g., made of nitinol) passively open to their greatest extent when relieved from radially inward compression, the lattice structure of the present invention is able to actively open according to the desire of the user surgeon implanting the prosthesis. As such, the opening performed by prior art self-expanding stent structures in endograft prosthesis are referred to herein as “passive opening” or “passive expansion”. In contrast thereto, the expansion performed by the inventive controllable, hinged, lattice structure of the present invention for the disclosed endograft prostheses is referred to herein as “active control” or “active expansion” because it can be actively controlled in both the expansion and contraction directions according to the desire of the user. This is further in contrast to expansion of stent structures using balloon, which case is referred to as “balloon opening” or “balloon expansion” because it occurs only in one direction (expansion) without any ability to contract actively. The single embodiment of the jack screw shown in FIGS. **14A** and **14B** can be replicated any number of times about the circumference of the lattice structure **2100**.

[0102] In a non-illustrated alternative to the configuration of the system shown in FIG. **14B**, the configuration shown

in FIGS. 10A to 11B can be incorporated into the system of FIGS. 14A and 14B to create a hybrid system. The circumferential assembly 1020 can be positioned at the proximal end of the endograft and action of the circumferential loop 1035 within the proximal cuff 1010, can be used to expand and contract the latticework 2100.

[0103] FIG. 15A is a lateral view of an exemplary embodiment of an adjustable vascular cannula 1230 according to the present invention. As shown in FIG. 15A, such an adjustable vascular cannula 1230 is a generally tubular structure with external cannula walls 1235 defining a cannula lumen 1240, and comprises a port end 1245, a cannula body 1250, and a cannula tip 1255. As further shown in FIG. 15A, the cannula body 1250 is further provided with a delivery recess 1260 in its external wall structure at or near the junction of the cannula tip 1255. Further still, the adjustable vascular cannula 1230 of FIG. 15A comprises an adjustable seal device 1265 attached to an adjustment member 1025 such as a torque wire that extends beyond the port end 1245 of the adjustable vascular cannula 1230 as shown in FIG. 15B. The adjustment member 1025 may course through the cannula lumen 1240, or it may course through an accessory lumen (not shown in FIGS. 15A or 15B) within the cannula wall 1235 substantially parallel to the cannula lumen 1240, or it may course externally to the adjustable vascular cannula 1230 as shown partially within and partially outside the lumen 1240 in FIG. 15B. When in a non-deployed state, as shown in FIG. 15B, the adjustable seal device 1265 is substantially flush with the outer diameter of the cannula walls 1235 within the delivery recess 1260 of the cannula body 1250.

[0104] FIG. 15C shows the adjustable seal device 1265 in a deployed state, which is the result of torque applied externally to the adjustment member 1025 by a user. As shown in FIG. 15C, the adjustable seal device 1265 further comprises a hinged adjustable latticework 1270 covered by a sealing cuff 1275 which is constructed of a distensible material. The adjustment member 1025 terminates, for example, in a circumferential loop 1035 within the sealing cuff 1275, where it may be further covered by a compressible foam gasket 1140. The adjustment member 1025 may further pass through a locking mechanism 1050 as disclosed elsewhere herein which serves to regulate the torque applied to the circumferential loop 1035. The hinged adjustable latticework 1270 may further be provided with one or more retention tines 1130, 1165, which are radially displaced from the terminal aspect of the hinged adjustable latticework 1270, and which are enclosed within and covered by the compressible foam gasket 1140 when the adjustable seal device 1265 is not distended. When torque is applied to the adjustment member 1025 by a user, the diameter of the circumferential loop 1035 is increased, displacing the hinged adjustable latticework 1270 as shown in FIG. 15C until the compressible foam gasket 1140 and the sealing cuff 1275 is able to firmly engage the inner wall 1190 of a recipient blood vessel 1175. A slight additional amount of torque applied to the adjustment member 1025 is, then, sufficient to compress the compressible foam gasket 1140 and allow the retention tines 1130, 1165 to engage the wall 1190 of the recipient blood vessel 1175, thus preventing slippage of the cannula during use. In various exemplary embodiments of the present invention, the retention tines 1130, 1165 may be provided to engage the vessel wall 1190 in a substantially straight manner or at angles varying from

about 1 degree to about 179 degrees. The retention tines 1130, 1165 may be angled axially or longitudinally in various embodiments according to the present invention. After the use of the cannula is completed, the torque of the adjustment member 1025 may be reversed, collapsing the adjustable seal device 1165, and allowing the compressible foam gasket 1140 to re-expand, thus withdrawing the retention tines 1165 from the vessel wall 1175 and covering the retention tines 1165 to allow atraumatic cannula withdrawal.

[0105] Although the foregoing embodiments of the present invention have been described in some detail by way of illustration and example for purposes of clarity and understanding, it will be apparent to those skilled in the art that certain changes and modifications may be practiced within the spirit and scope of the present invention. Therefore, the description and examples presented herein should not be construed to limit the scope of the present invention, the features of which are set forth in the appended claims.

[0106] The foregoing description and accompanying drawings illustrate the principles, exemplary embodiments, and modes of operation of the invention. However, the invention should not be construed as being limited to the particular embodiments discussed above. Additional variations of the embodiments discussed above will be appreciated by those skilled in the art and the above-described embodiments should be regarded as illustrative rather than restrictive. Accordingly, it should be appreciated that variations to those embodiments can be made by those skilled in the art without departing from the scope of the invention as defined by the following claims.

What is claimed is:

1. An assembly comprising:

an implant body having a lattice structure, an adjustment member, and a control device, wherein the lattice structure is circumferentially expandable and contractible, the adjustment member and the control device are coupled to the lattice structure, and actuating the adjustment member results in circumferential expansion or contraction of the lattice structure; and

a delivery apparatus having a rotatable adjustment tool and a locking mechanism comprising a controllable catch coupled to an end portion of the adjustment tool, wherein the adjustment tool is configured to actuate the adjustment member of the implant body upon rotation of the adjustment tool, and the controllable catch of the locking mechanism is configured to selectively couple the adjustment tool to the adjustment member of the implant body.

2. The assembly of claim 1, wherein the controllable catch of the locking mechanism is movable between an engaged configuration and a disengaged configuration, wherein the controllable catch of the locking mechanism engages an adjustment member catch of the adjustment member such that axial movement of the adjustment tool results in corresponding axial movement of the adjustment member when the locking mechanism is in the engaged configuration, and wherein the controllable catch of the locking member disengages the adjustment member catch of the adjustment member such that the adjustment tool can move axially relative to the adjustment member when the locking mechanism is in the disengaged configuration.

3. The assembly of claim 2, wherein when the controllable catch of the locking mechanism is in the engaged configuration, the controllable catch of the locking mechanism

engages the adjustment member catch of the adjustment member such that rotational movement of the adjustment tool results in corresponding rotational movement of the adjustment member relative to the lattice structure, rotating the adjustment tool in a first direction actuates the adjustment member and radially expands the lattice structure, and rotating the adjustment tool in a second direction actuates the adjustment member and radially contracts the lattice structure.

4. The assembly of claim 1, wherein the adjustment member is a micro-threaded cable.

5. The assembly of claim 4, wherein the control device is in the form of a microcylinder comprising a threaded aspect configured to receive the adjustment member.

6. The assembly of claim 1, wherein the adjustment member comprises a jack screw.

7. The assembly of claim 1, wherein the adjustment member comprises a plurality of jack screws distributed circumferentially around the lattice structure.

8. The assembly of claim 7, wherein the shaft comprises a plurality of shafts, and the locking mechanism comprises a plurality of locking mechanisms, the locking mechanisms are coupled to respective shafts, and the locking mechanisms are configured for selectively coupling the shafts to respective jack screws.

9. An assembly comprising:

a lattice structure, wherein the lattice structure is circumferentially expandable and contractible;

a torque wire, wherein the torque wire is coupled to the lattice structure at two points longitudinally separate from one another on the lattice structure and configured for circumferentially expanding and contracting the lattice structure; and

an adjustment tool, wherein when the adjustment tool is rotated, the torque wire correspondingly rotates to expand or contract the lattice structure.

10. The assembly of claim 9, further comprising a smooth-bored collar through which the torque wire extends.

11. The assembly of claim 9, wherein the torque wire has exterior threads that correspond to threaded bores of one of the two points.

12. The assembly of claim 9, wherein when the torque wire is rotated, the two points of the lattice structure either approach one another or retreat from one another.

13. An assembly comprising: a radially expandable and contractible stent; a jack screw coupled to the stent, wherein rotating the jack screw in a first direction relative to the stent radially expands the stent, and rotating the jack screw in a second direction relative to the stent radially contracts the stent; and

an adjustment tool configured for rotating the jack screw.

14. The assembly of claim 13, wherein the stent has a plurality of bores longitudinally separate from each other, wherein at least one bore of the plurality of bores is a threaded bore with threads that correspond to exterior threads of the jack screw, wherein at least one other bore of the plurality of bores is a non-threaded bore, and wherein the jack screw extends longitudinally through the threaded bore and the non-threaded bore.

15. The assembly of claim 13, wherein the adjustment tool has a jack screw disposed within at least some of the threaded bores and have exterior threads corresponding to interior threads of the threaded bores, wherein rotating the jack screw in a first direction relative to the threaded bores moves the threaded bores axially closer together, and rotating the jack screw in a second direction relative to the threaded bores moves the threaded bores axially apart.

16. The assembly of claim 15, wherein the locking mechanism comprises a catch member that engages the jack screw when the locking mechanism is in the lock configuration and that disengages the jack screw when the locking mechanism is in the release configuration.

17. The assembly of claim 16, wherein the adjustment tool comprises a plurality of adjustment tools, the shaft comprises a plurality of shafts, and the locking mechanism comprises a plurality of locking mechanisms, wherein each of the locking mechanisms is configured to releasably couple one of the shafts to a respective adjustment tool.

18. The assembly of claim 17, wherein the jack screw of each adjustment tool is circumferentially-spaced relative to an adjacent jack screw.

19. The assembly of claim 13, wherein the adjustment tool is a cable.

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